Tire Fire Contingency Plan - Toxicology Aspects

Prepared for

Tacoma-Pierce County Health Department

Tacoma, Washington

Tacoma-Pierce County Health Department

September 1994

Adolfson Associates, Inc.
In association with
Kim Coble

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PREFACE

Safe, practical management of waste tires is a problem facing almost every community in our country. An estimated 164 million tires per year are discarded across the United States, the vast majority of which usually end up along roadsides, in landfills or stockpiled. Each of these disposal methods has its own set of disadvantages; however, by far the greatest threat from any of these options is the potential for stockpiled tires to catch on fire. A tire fire can result in significant environmental damage and poses a health risk to responding fire fighters and to nearby residents. Additional health risks exist due to other unknown sources of contamination associated with tire piles (e.g., wheels, lead weights, hazardous materials, etc.).

The best method to reduce threats from a tire fire to the environment and human health is to be prepared. Plans regarding tire fire management from a fire fighter's perspective have been written and can be accessed through local emergency service agencies and fire departments. This document takes response planning one step further, and assists health agencies in responding to the environmental health issues associated with a tire fire.

This document is divided into five sections, each of which explains the environmental health issues related to a typical tire fire incident. Section I reviews the current data on tire fire emissions and selects a subset of target chemicals that could be sampled during a tire fire. A discussion of exposure pathways is also included. Section II describes air monitoring methods and specifically discusses sampling methods for the target chemicals. Section III focuses on surface water and ground water contamination that can result from a tire fire and describes the sampling methods that may be used to assess such contamination. Section IV addresses personal protective equipment and safety planning. Section V provides a general discussion of how to effectively communicate with the media and the public about the risks associated with a tire fire incident. It also gives three sample health fact sheets (targeted toward the media, public, and fire fighters) developed for a hypothetical tire fire. These fact sheets could be used as a template and adapted for an actual tire fire.

Wherever tires are stockpiled, whether there are several hundred or several million, the threat of a tire fire exists. A health agency must be prepared for a tire fire in order to minimize the environmental health risks. This document will assist health agency planning and response to the environmental health issues associated with a tire fire. This document is not meant to be a final contingency plan for every tire fire, rather it provides a framework that can be adapted for a site specific plan as needed.

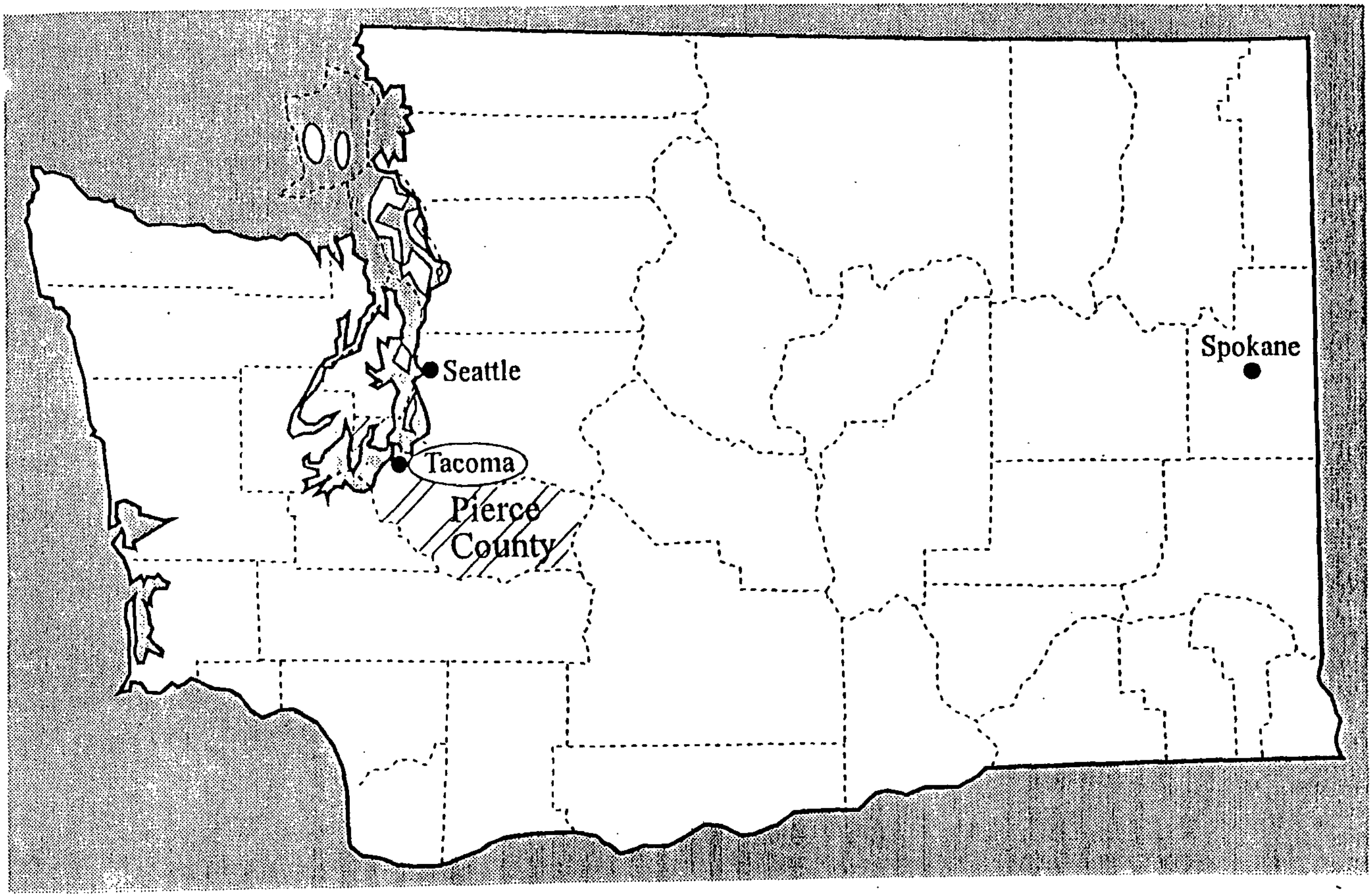


Figure i: Tacoma, Pierce County Location Map

EXECUTIVE SUMMARY

The potential environmental health issues that could result from a tire fire are far reaching. The best method for a health agency to address these issues is to be prepared. Agency personnel should have knowledge of the chemicals which are released from the fire and their toxicity, the exposure pathways, what chemicals to sample and how to sample for them, how to protect the health of the workers on the site and finally, how to effectively communicate to the media and nearby residents about the potential risks of the situation. Tire piles should be assessed prior to a fire to determine additional sources of contamination within the tire pile (e.g., wheels, hazardous materials, etc.). This document was written to assist health agencies in responding to these environmental health issues associated with a tire fire.

The target chemical selection process identified 38 chemicals that should be considered for air monitoring during a tire fire. The selection was based upon toxicity and expected concentration. These chemicals were primarily hydrocarbons, metals and inorganic gases and vapors. There is a wide range of adverse health effects that could result from exposure to these chemicals. Generally speaking, the health effects can be quite severe and include cancer, central nervous system depression, respiratory effects and irritation of the skin, eyes and mucous membranes.

The primary route of exposure at a tire fire is inhalation of gases, vapors and particulates. The highest levels of chemicals occur close to the fire and pose the greatest risk to workers on-site. Proper personal protective equipment for workers will eliminate exposure to the chemicals by providing respiratory and dermal protection. Level B protective equipment has been used at previous tire fires and is recommended because it protects both the respiratory system and the skin and yet, is less cumbersome than level A. Using level B versus level A also will help protect workers from heat stress.

The risk of exposure to nearby residents can be assessed through air, water and soil monitoring. There are two types of approaches for identifying and/or quantifying airborne contaminants: direct-reading instruments, and laboratory analysis of air samples. Both approaches should be used to assess the levels of chemicals present. Air samples should be collected both near the fire and away from the fire in the nearby communities. Data from previous tire fires have indicated that chemical levels detected at the edge of the site boundaries generally are low, and that the risk to the nearby communities is minimal. However, risk assessments should be continually verified through air monitoring throughout the tire fire event. Action levels for interpreting the results include threshold limit values and reference concentrations. Community actions that could be implemented are continual air monitoring, issuing a warning, restricting outdoor activities, and evacuation.

Water and soil provide secondary exposure pathways primarily through ingestion of contaminated water, soil particles, plants and potentially, fish. Water contamination can occur from oil and water run-off to surface waters and/or through seepage into ground water. Samples of on-site surface waters, runoff from fire fighting efforts, as well as samples from nearby surface water bodies should be collected. Ground water sampling is recommended in

the event of extensive surface water and soil contamination. However, if contaminated soil is removed quickly, ground water sampling may not be necessary. Soil samples collected from areas of run-off and nearby stream beds may be useful in determining the extent of contamination, what contaminants may have entered the ground water and air, and whether clean-up and disposal of the soil is necessary. Sample collection locations and the frequency of sample collection should be determined on a site-specific basis.

Chemical detection in different media and assuring the health and safety of workers and nearby residents are major challenges during a tire fire incident; however, often the greater challenge is effectively communicating with the media and the citizens about the potential risks of the situation. Open and honest communication must start immediately. Be prepared to answer who, what, where, when and how questions the press will have. Keep the information simple and concise. In terms of communicating with the public, one of the best methods to control their "outrage" is by involving them in the decision process from the beginning and communicating with them regularly through fact sheets or other methods.

The environmental health risks from a tire fire are both significant and controllable. The risks can be controlled using proper sampling and analyses to identify and quantify the chemicals, wearing the correct personal protective equipment, and effectively communicating the risk to the media and the public. The best method to minimize environmental health risks is by planning ahead.

I. CHEMICAL COMPOUND PROFILE

INTRODUCTION

The purpose of this section is to identify the chemicals which are released during a tire fire and select a subset of chemicals that should be monitored during a tire fire. The selection of "target chemicals" was based upon potential health effects (the toxicity of the chemical) and the amount emitted during a tire fire (the concentration of the chemical). Chemicals found in smoke and pyrolytic oil were identified. In addition, the health effects and the exposure routes for the target chemicals were assessed.

APPROACH

Initially, the first approach to chemical selection was to review the emissions data collected from tire fires and determine which chemicals were of most concern. A review of published studies and reports discussing the emissions from tire fires indicated that a variety of collection and analytical methods were used as well as differing levels of quality control. Based upon the variations in sampling methods, the data sets were not easily comparable. After agreement from Tacoma Pierce County Health Department (TPCHD), a second approach to this task was taken.

The second approach was to use two existing data sets with consistent methods and reliable quality control procedures. One set is the air monitoring data collected by the Environmental Protection Agency (EPA) at nine tire fire locations. The EPA data came from emissions data from tire fires in Wisconsin, Washington, Virginia, Arkansas, Colorado, North Carolina, New York, Pennsylvania, and Utah. The other set is from a test burn which identified and quantified organic and inorganic emissions produced during the burning of scrap tires. Data were collected from test burns of both chunked and shredded tires. The chunk tires were cut into pieces 1/4 to 1/6 the size of an automobile-sized tire. The shredded tires measured 2 inches by 2 inches. For the purposes of this report, the chunk data were the only data used, as it was concluded that the chunk tires were more representative of the type of material that would be found at a tire pile fire. These data were collected under laboratory conditions and are a good indication of the emissions within close proximity to the fire. These data are given in a report called "Characterization of Emissions from the Simulated Open Burning of Scrap Tires" by Acurex Corp. and are referred to in this document as the Acurex data.

The Acurex and EPA data sets were evaluated and target chemicals were selected based upon toxicity and concentration. Figure 1 gives a flow chart for target chemical selection.

The first step was simply that the chemical had to be in either the EPA or the Acurex data set and have a detectable level of emission. The second step was to evaluate whether or not the chemical was a suspected or confirmed human carcinogen. If it was, it became a target chemical, regardless of the recorded emission value. Carcinogenicity was assessed through the Health Effects Assessment Summary Tables produced by EPA and the Threshold Limit Values

FIGURE 1: TARGET CHEMICAL SELECTION - FLOW CHART

Eliminated ← NO ← | Is chemical listed in Acurex and/or EPA Data base?

↓

YES

↓

Is chemical human carcinogen or suspected human carcinogen? | → YES → target chemical

↓

NO/ND

↓

Is monitored value greater than 33% of the TLV? | → YES → target chemical

↓

NO/ND

↓

Is monitored value greater than the SC RfC or the C RfC? | → YES → target chemical

↓

NO/ND

↓

Eliminated ← NO ← | Is monitored value in top 25% of the database and not previously selected or eliminated?

↓

YES

↓

Eliminated ← NO/ND ← | Is chemical of high human toxicity and high concentration? | → YES → target chemical

TLV = Threshold Limit Value
SC RfC = Subchronic Inhalation Reference Concentration
C RfC = Chronic Inhalation Reference Concentrations
ND = No Data

list produced by the American Conference of Governmental Industrial Hygienists. The one exception to this is that some of the carcinogenic polyaromatic hydrocarbons (PAHs) that were in low concentration were not singled out. Coal tar pitch volatiles were used as representative of the PAH class of chemicals.

The third step was to determine if a threshold limit value (TLV) exists for the chemical. The TLVs refer to airborne concentrations and represent conditions under which it is believed that nearly all workers may be repeatedly exposed eight hours a day, five days a week, without adverse health effects. TLVs are based on the best available information from industrial experience, and experimental studies on humans and animals, and when possible, all three. It should be noted that TLVs are guidelines or recommendations, not standards for determining safe or unsafe conditions. If the detected value was greater than 33% of the TLV, it became a target chemical. Thirty-three percent of the TLV was used because the TLV is based upon a typical work week (8 hours/day, 40 hours/week) and in the case of a tire fire, exposure could be 24 hours per day. In cases where several types of a chemical were listed in the TLV book (e.g., nickel, nickel carbonyl, nickel sulfide), a specific chemical was selected by determining which was the most likely one to come from a tire fire and/or which had the lowest TLV. In this report, TLVs were one of several indicators used to select target chemicals.

The fourth step was to compare the detected value to the subchronic (SC) and chronic (C) inhalation reference concentrations (RfC). The RfC is an estimate of the daily exposure to the human population that is likely to be without an appreciable risk of adverse effects during a portion of the lifetime (subchronic) or during the lifetime (chronic). Sensitive populations are taken into account. The RfC is based upon the No Observable Adverse Effect Level (NOAEL) or the Lowest Observable Adverse Effect Level (LOAEL), an uncertainty factor, and a modifying factor. The RfCs are for continuous, 24 hours/day exposure, and are used as reference points for gauging the potential effects of exposures. Usually, doses that are less than the RfC are not likely to be associated with health risks. As the frequency of exposures exceeding the RfC increases, and as the size of the excess increases, the probability increases that adverse health effects may be observed in a human population. Like the TLV, the RfC is not meant to be a standard or give a definitive statement about safe and unsafe concentrations. However, it is appropriate to use it in this case when choosing target chemicals.

The fifth step was to list the chemicals in order of decreasing value and to take the top 25% of the chemicals from each data set. If a chemical had not already been targeted as a carcinogen or one that exceeded the RfC or 33% of the TLV, a review of the chemical was conducted. The toxicity and the concentration were evaluated and a decision was made as to whether or not it should be selected as a target chemical. If no information could be found, the chemical was not included as a target chemical.

It should be noted that if a range was reported in the EPA or Acurex data set, the maximum value given was the value used during the selection process. It should also be noted that this approach was not meant to provide an all inclusive description of the emissions from a tire fire, rather, it was used to select the most likely chemicals that are released from tire fires. As a check, the list of target chemicals was compared to actual tire fire data from a variety of other reports and they were consistent with the findings from other reports. (Turpin, 1984; Allen, 1984; Turpin, 1984)

FINDINGS

Tire Fire Composition

The EPA's report entitled *Burning Tires for Fuel and Tire Pyrolysis: Air Implications* (EPA, 1991) shows the chemical products of burned tires as follows:

64-87%	carbon
5 - 7%	hydrogen
2 - 5%	oxygen
<1%	nitrogen
1 - 2%	sulfur
3 - 25%	ash
22 separate metals	

Ranges are reported because the amount varies depending on the type of tire.

A summary by the Goodyear Tire and Rubber Company indicated tire composition to be as follows: synthetic rubber, natural rubber, sulfur and compounds, clay phenolic resin, oil, fabric, petroleum waxes, pigments, carbon black, fatty acids, inert materials and wire.

Target Chemical Selection

The Acurex data set contained 82 chemicals and the EPA data set contained 68 chemicals. A list of all the chemicals and their maximum reported values is shown in Table 1. From these 150 chemicals, 38 were selected as target chemicals. Table 2 gives the list of target chemicals and the criterion for selection. This selection was based upon the decision process shown in Figure 1.

Suspected or Confirmed Human Carcinogens

Table 3 lists 26 chemicals that are suspected or confirmed human carcinogens and their maximum reported value. All suspected or confirmed carcinogens were chosen as target chemicals regardless of their value. Cancer is considered a result of chronic exposure. It is unlikely that there would be a chronic exposure to smoke from a tire fire, and therefore, unlikely that someone would contract cancer from a tire fire. However, extra caution in protecting human health is being taken by including the carcinogens as target chemicals.

Many of the chemicals that are detected are polycyclic aromatic hydrocarbons (PAHs). PAHs are a class of compounds consisting of various arrangements and substitutions of multiple benzene rings. They are formed during the incomplete burning of coal, oil and gas, garbage,

TABLE 1: (cont.) CHEMICALS FROM ACUREX AND EPA DATASETS AND MAXIMUM REPORTED VALUES

SOURCE	*CHEMICAL*	*VALUE (mg/m3)*
Acu	Acenaphthene	1.027
Acu	Acenaphthylene	0.897
EPA	Acetone	0.3700
Acu	Aluminum	0.007
Acu	Anthracene	0.044
EPA	Anthracene	0.0330
Acu	Antimony	0.007
Acu	Arsenic	0.0002
Acu	Barium	0.0035
Acu	Benz(a) Anthracene	0.226
EPA	Benz(a)Anthracene	0.0018
Acu	Benzaldehyde	0.566
EPA	Benzene	10.59
Acu	Benzene	3.872
Acu	Benzo(a) Pyrene	0.481
EPA	Benzo(a) Pyrene	0.0130
Acu	Benzo(b) Fluoranthene	0.344
Acu	Benzo(g,h,i)Perylene	0.441
Acu	Benzo(k) Fluoranthene	0.328
Acu	Benzodiazine	0.033
Acu	Benzofuran	0.169
Acu	Benzothiophene	0.03
EPA	Benzylchloride	0.0190
Acu	Biphenyl	0.29
Acu	Biphenyl Methyl, 1,1-	0.025
EPA	Bromide	94.00
EPA	Bromochloromethane	1.1360
Acu	Butadiene	0.314
EPA	Butyl Toluene, 4-Tert	0.1310
Acu	Calcium	0.014
EPA	Carbon Monoxide	100.00*
EPA	Carbon Tetrachloride	0.0082*
EPA	Chloride	83.00
EPA	Chloroform	2.0580
Acu	Chromium	0.0058*
EPA	Chrysene	0.4460

TABLE 1: (cont.) CHEMICALS FROM ACUREX AND EPA DATASETS AND MAXIMUM REPORTED VALUES

SOURCE	*CHEMICAL*	*VALUE (mg/m3)*
Acu	Chrysene	0.368
EPA	Coal Tar Pitch	4.2180
Acu	Copper	0.0007
EPA	Cumene	0.0940
Acu	Cyanobenzene	0.305
EPA	Cyclohexane	0.0630
Acu	Cyclopentadiene	0.273
Acu	Dibenz(a,h,)Anthracene	0.007
EPA	Dichlorobenzene, 1,2-	0.0696
EPA	Dichlorobenzene, 1,4-	0.1187
EPA	Dichloropropane, 1,2-	0.0350
Acu	Dihydroindene	0.05
Acu	Dimethyl Benzene (Xylene)	0.789
Acu	Dimethyl Hexadiene	0.038
Acu	Dimethyl Naphthalene	0.069
Acu	Dimethyldihydro Indene	0.029
Acu	Ethenyl Benzene	1.205
Acu	Ethenyl Cyclohexene	0.038
Acu	Ethenyl,Dimethyl Benzene	0.067
Acu	Ethenylmethyl Benzene	0.315
Acu	Ethyl Benzene	0.537
EPA	Ethyl Benzene	0.1554
EPA	Ethyl Toluene	0.0110*
Acu	Ethyl,Methyl Benzene	0.338
EPA	Ethylene Dichloride	0.3230
EPA	Ethyltoluene, Meta-	0.1180*
Acu	Ethynyl Benzene	0.974
Acu	Ethynyl,Methyl Benzene	1.347
Acu	Fluoranthene	0.473
EPA	Fluoranthene	0.0040
Acu	Fluorene	0.277
EPA	Fluorene	0.0260
EPA	Fluoride	32.00
Acu	Fluorene, 1H	0.288
Acu	Heptadiene	0.031
EPA	Heptane	0.0200

TABLE 1: (cont.) CHEMICALS FROM ACUREX AND EPA DATASETS AND MAXIMUM REPORTED VALUES

SOURCE	*CHEMICAL*	*VALUE (mg/m3)*
EPA	Heptane, N-	0.0830
EPA	Hexachloroethane	0.2980
Acu	Hexahydro Azepinone	0.126
EPA	Hexane	0.2100
EPA	Hexane, N-	0.1580
EPA	Hexene	0.0200
EPA	Hydrobromic Acid	0.2550
EPA	Hydrochloric Acid	4.0000
EPA	Hydrofluoric Acid	0.2700
Acu	Indene	0.602
Acu	Indeno(1,2,3-CD)Pyrene	0.312
EPA	Iron	0.0140
Acu	Iron	0.0308
Acu	Isocyano Benzene	1.33
Acu	Isocyano Naphthalene	0.021
EPA	Lead	0.0110
Acu	Lead	0.0007
Acu	Limonene	0.094
Acu	Magnesium	0.0021
Acu	Methyl Benzene (Toluene)	1.556
Acu	Methyl Cyclohexene	0.03
EPA	Methyl Ethyl Ketone	0.0580
Acu	Methyl Hexadiene	0.095
Acu	Methyl Indene	0.231
Acu	Methyl Naphthalene	0.372
Acu	Methyl Thiophene	0.02
Acu	Methyl,Ethenyl Benzene	0.057
Acu	Methyl,Methylethenyl Benzene	0.103
Acu	Methyl,Methylethyl Benzene	0.215
EPA	Methylene Chloride	0.0606*
Acu	Methylene Indene	0.074
Acu	Methylethyl Benzene	0.161
EPA	Methylstyrene, 3-	0.0960
EPA	Methylstyrene, 4-	0.0500
EPA	Methylstyrene, Alpha-	0.0500
Acu	Naphthalene	2.953

TABLE 1: (cont.) CHEMICALS FROM ACUREX AND EPA DATASETS AND MAXIMUM REPORTED VALUES

SOURCE	*CHEMICAL*	*VALUE (mg/m3)*
EPA	Naphthalene	1.3200
Acu	Nickel	0.007
EPA	Nitrate-N	220.00
EPA	Nitric Acid	0.2550
EPA	Octane, N-	0.0850
EPA	Orthophosphate	280.00
Acu	Pentadiene	0.146
EPA	Pentane	0.6600
EPA	Pentane, N-	0.2960
Acu	Phenanthrene	0.201
EPA	Phenanthrene	0.0340
Acu	Phenol	0.473
EPA	Phosphoric Acid	0.2650
Acu	Propenyl Naphthalene	0.053
Acu	Propyl Benzene	0.086
Acu	Pyrene	0.192
EPA	Pyrene	0.0001
EPA	Pyrylene	2.6230
Acu	Selenium	0.0002
Acu	Silicon	0.0944
Acu	Sodium	0.0175
EPA	Styrene	5.4100
Acu	Styrene	0.795
EPA	Sulfate	230.00
EPA	Sulfur Dioxide	1.0320*
EPA	Sulfuric Acid	0.7900
EPA	Thiophene	0.3000
Acu	Thiophene	0.064
Acu	Titanium	0.0175
EPA	Toluene	6.7000
EPA	Trichloroethane, 1,1,2-	0.000545*
EPA	Trichloroethane, 1,1,1-	0.0140*
EPA	Trichloroethylene	0.2900*
Acu	Trichlorofluoromethane	1.003
EPA	Trichlorofluoromethane	0.0090*
Acu	Trimethyl Benzene	0.258

TABLE 1: (cont.) CHEMICALS FROM ACUREX AND EPA DATASETS AND MAXIMUM REPORTED VALUES

SOURCE	*CHEMICAL*	*VALUE (mg/m3)*
Acu	Vanadium	0.0175
EPA	Xylene, M,P	131.00
EPA	Xylene, M-	0.1950*
EPA	Xylene, O-	1564.00
EPA	Zinc	0.0130
Acu	Zinc	0.1259

* = value is in ppm
Acu = Acurex Dataset
EPA = EPA Dataset

TABLE 2: TARGET CHEMICALS BY CRITERIA

	CA	TLV	SC RfC	C RfC
Acenaphthene	x			
Acenaphthylene	x			
Arsenic	x			
Barium				x
Benz(a)Anthracene	x			
Benzene	x			
Benzo(a)Pyrene	x			
Benzo(b)Fluoranthene	x			
Benzylchloride	x			
Butadiene	x			
Carbon Monoxide		x		
Carbon Tetrachloride	x			
Chloride			x	x
Chloroform	x			
Chromium	x			
Chrysene	x			
Coal Tar Pitch	x	x		
Cumene			x	x
1,2-Dichloropropane	x		x	x
Dibenz(a,h)Anthracene	x			
Ethylene Dichloride	x			
Fluoride		x		
Hexachloroethane	x			
Hexane			x	x
Lead	x			
Methylene Chloride	x			
Nickel	x			
Phenol	x			
Styrene	x			x
Sulfur Dioxide		x		
Sulfuric Acid		x		x
Toluene (methyl benzene)			x	x
Trichloroethane, 1,1,2-	x			
Trichloroethylene	x			
Vanadium		x		
Xylene, 0-		x		
Zinc	x	x		

CA = Suspected or Confirmed Human Carcinogen
TLV = Reported Value is 33% of Threshold Limit Value
SC RfC = Sub Chronic Inhalation Reference Concentration
C RfC = Chronic Inhalation Reference Concentration

TABLE 3: CHEMICALS THAT ARE SUSPECTED OR CONFIRMED HUMAN CARCINOGENS AND MAXIMUM REPORTED VALUE

SOURCE	CHEMICAL	VALUE (mg/m3)
Acu	Acenaphthene	1.027
Acu	Acenaphthylene	0.897
Acu	Arsenic	0.0002
Acu	Benz (a) Anthracene	0.226
EPA	Benzene	10.59
Acu	Benzene	3.872
Acu	Benzo(a)Pyrene	0.481
EPA	Benzo(a)Pyrene	0.013
Acu	Benzo(b)Fluoranthene	0.344
EPA	Benzylchloride	0.019
Acu	Butadiene	0.314
EPA	Carbon Tetrachloride	0.0082*
EPA	Chloroform	2.058
Acu	Chromium	0.0058*
EPA	Chrysene	0.446
Acu	Chrysene	0.368
EPA	Coal Tar Pitch	4.218
Acu	Dibenz (a,h) Anthracene	0.007
EPA	Dichloropropane, 1,2-	0.035
EPA	Ethylene Dichloride	0.323
EPA	Hexachloroethane	0.298
Acu	Lead @ (inorganic dust)	0.0007
EPA	Lead@ (inorganic dust)	0.011
EPA	Methylene Chloride	0.0606*
Acu	Nickel @ (metal)	0.007
Acu	Phenol	0.473
EPA	Styrene	5.41
Acu	Styrene	0.795
EPA	Trichloroethane, 1,1,2-	0.0005*
EPA	Trichloroethylene	0.29*
Acu	Zinc @ (chromates)	0.1259
EPA	Zinc @ (chromates)	0.013

* = ppm
@ = specific chemical selected from TLV list in parenthesis

wood or other organic substances. PAHs can be man-made or occur naturally. They are found throughout the environment, almost always in groups of two or more. They can occur in the air either attached to dust particles or in the soil or sediment as solids. They can be found in substances such as crude oil, coal tar pitch, creosote, and road and roofing tar. Background levels of PAHs in the air are 0.02-1.2 mg/m^3 in rural areas and 0.15-19.3 mg/m^3 in urban areas. Most of the higher weight PAHs are suspected human carcinogens. Many of the chemicals from both the Acurex and EPA data are PAHs. Data from other tire fires, including data from two large tire fires, the Everett, Washington and Winchester, Virginia tire fires indicated PAHs were the primary category of chemicals.

Threshold Limit Values

The chemicals for which there is a TLV, the maximum reported value and the TLV are listed in Table 4. The last column is the reported value divided by the TLV. This column gives some indication of the value relative to the TLV. If the reported value was greater than 33% of the TLV it was selected as a target chemical. There were eight chemicals that were greater than 33% of the TLV. Five of the eight were significantly greater than the TLV, e.g., coal tar pitch exceeded the TLV by 2,109% and fluoride exceeded it by 1,280%. Zinc was reported in both the EPA and the Acurex data and was 1,259% greater than the TLV in the Acurex data and 130% greater using the EPA data. Carbon monoxide was 400% greater and O-xylene was 360% greater than the TLV. Sulfur dioxide, sulfuric acid and vanadium were below 100%, but greater than 33% with 52%, 79%, and 35% of the TLVs respectively.

It is not surprising that coal tar pitch was recorded as having the highest percentage, over 33% of the TLV. Coal tar pitch volatiles are a group of lower molecular weight polycyclic hydrocarbons (naphthalene, fluorene, anthracene, acridine, phenanthrene) that sublime into the air and result in an increase of higher molecular weight polycyclic hydrocarbons (benzo(a)pyrene, benz(a)anthracene), many of which are carcinogenic to humans. Coal tar pitch contains approximately 10% polycyclic aromatic hydrocarbons. It is used as a base for coatings and paints, for roofing and paving, and as a binder for carbon electrodes. As stated earlier, PAHs have been shown to be the primary class of chemicals found at tire fires.

It is not understood why fluoride was found in such high levels. It may be due to calcium fluoride which is used in steelmaking, and that as the steel belted tires burn, fluoride is released. The data did not indicate the specific fluoride compound present and therefore, it is difficult to speculate as to why the excessively high value is present.

The major use of zinc pigments and salts include rubber goods which would include tire manufacturing. This could explain the high level of zinc, however, once again, the specific zinc compound was not identified.

Carbon monoxide is one of the most widely encountered gases. It is a product of incomplete combustion when carbonaceous matter such as coal, natural gas, gasoline, oil and any other organic matter is burned. Carbon monoxide is consistently found in tire fire emissions and is considered a good compound for direct-reading monitoring at tire fires. (See Section II).

TABLE 4: (cont.) CHEMICALS THAT HAVE THRESHOLD LIMIT VALUES BY MAXIMUM REPORTED VALUE, TLV AND PERCENT TLV

SOURCE	*CHEMICAL*	*VALUE (mg/m3)*	*TLV (mg/m3)*	*%TLV*
Acu	Iron @ (oxide fume)	0.0308	5	0.62%
EPA	Iron @ (oxide fume)	0.0140	5	0.28%
EPA	Lead @ (inorganic dust)	0.0110	0.15	7.33%
Acu	Lead @ (inorganic dust)	0.0007	0.15	0.47%
Acu	Magnesium	0.0021	10	0.02%
EPA	Methyl Ethyl Ketone	0.0580	590	0.01%
EPA	Methylene Chloride	0.0606*	50*	0.12%
EPA	Methylstyrene, Alpha-	0.0500	242	0.02%
Acu	Napthalene	2.953	52	5.68%
EPA	Napthalene	1.3200	52	2.54%
Acu	Nickel @ (metal)	0.007	1	0.70%
EPA	Nitric Acid	0.2550	5.2	4.90%
EPA	Octane, N-	0.0850	1400	0.01%
EPA	Pentane	0.6600	1770	0.04%
EPA	Pentane, N-	0.2960	1770	0.02%
Acu	Phenol	0.473	19	2.49%
EPA	Phosphoric Acid	0.2650	1	26.50%
Acu	Selenium	0.0002	0.2	0.10%
Acu	Silicon	0.0944	10	0.94%
Acu	Sodium @ (hydroxide)	0.0175	2	0.88%
EPA	Styrene	5.4100	213	2.54%
Acu	Styrene	0.795	213	0.37%
EPA	**Sulfur Dioxide**	**1.0320***	**2***	**52.00%**
EPA	**Sulfuric Acid**	**0.7900**	**1**	**79.00%**
Acu	Titanium @ (dioxide)	0.0175	10	0.18%
EPA	Toluene	6.700	188	3.56%
EPA	Trichloroethane, 1,1,2-	0.000545*	10*	0.01%
EPA	Trichloroethane, 1,1,1-	0.0140*	350*	0.00%
EPA	Trichloroethylene	0.2900*	50*	0.58%
Acu	Trichlorofluoromethane	1.003	5620	0.02%
EPA	Trichlorofluoromethane	0.0090*	1000*	0.00%
Acu	Trimethyl Benzene	0.258	123	0.21%
Acu	**Vanadium @ (pentoxide)**	**0.0175**	**0.05**	**35.00%**
EPA	Xylene, M,P-	131.00	434	30.18%
EPA	Xylene, M-	0.1950*	100*	0.20%

TABLE 4: (cont.) CHEMICALS THAT HAVE THRESHOLD LIMIT VALUES BY MAXIMUM REPORTED VALUE, TLV AND PERCENT TLV

SOURCE	*CHEMICAL*	*VALUE (mg/m3)*	*TLV (mg/m3)*	*%TLV*
Acu	Aluminum	0.007	10	0.07%
Acu	Antimony	0.007	0.5	1.40%
Acu	Arsenic	0.0002	0.2	0.10%
Acu	Barium	0.0035	0.5	0.70%
EPA	Benzene	10.59	32	33.09%
Acu	Benzene	3.872	32	12.10%
EPA	Benzylchloride	0.0190	5.2	0.37%
Acu	Biphenyl	0.29	1.3	22.31%
EPA	Bromochloromethane	1.1360	1060	0.11%
Acu	Butadiene	0.314	22	1.43%
EPA	Butyl Toluene, 4-Tert-	0.1310	61	0.21%
Acu	Calcium @ (carbonate)	0.014	10	0.14%
EPA	**Carbon Monoxide**	**100.0000***	**25***	**400.00%**
EPA	Carbon Tetrachloride	0.0082*	5*	0.17%
EPA	Chloroform	2.0580	49	4.20%
Acu	Chromium	0.0058*	0.05*	11.60%
EPA	**Coal Tar Pitch**	**4.2180**	**0.2**	**2109%**
Acu	Copper	0.0007	1	0.07%
EPA	Cumene	0.0940	246	0.04%
EPA	Cyclohexane	0.0630	1030	0.01%
Acu	Cyclopentadiene	0.273	203	0.13%
EPA	Dichlorobenzene, 1,2-	0.0696	150	0.05%
EPA	Dichlorobenzene, 1,4-	0.1187	451	0.03%
EPA	Dichloropropane, 1,2-	0.0350	347	0.01%
Acu	Dimethyl Benzene (xylene)	0.789	434	0.18%
Acu	Ethyl Benzene	0.537	434	0.12%
EPA	Ethyl Benzene	0.1554	434	0.04%
EPA	Ethylene Dichloride	0.3230	40	0.81%
EPA	**Fluoride**	**32.0000**	**2.5**	**1280%**
EPA	Heptane	0.0200	1640	0.00%
EPA	Heptane, N-	0.0830	1640	0.01%
EPA	Hexachloroethane	0.2980	9.7	3.07%
EPA	Hexane	0.2100	176	0.12%
EPA	Hexane, N-	0.1580	176	0.09%
Acu	Indene	0.602	48	1.25%

TABLE 4: (cont.) CHEMICALS THAT HAVE THRESHOLD LIMIT VALUES BY MAXIMUM REPORTED VALUE, TLV AND PERCENT TLV

SOURCE	*CHEMICAL*	*VALUE (mg/m3)*	*TLV (mg/m3)*	*%TLV*
EPA	**Xylene, O-**	**1564.00**	**434**	**360.37%**
Acu	**Zinc @ (chromate)**	**0.1259**	**0.01**	**1259.00%**
EPA	**Zinc @ (chromate)**	**0.0130**	**0.01**	**130.00%**

* = values in ppm
@ = specific chemical selected from TLV list in parenthesis
Bolded = values exceed 33% of TLV

Xylene is a petrochemical and is used extensively as a solvent, in the synthesis of numerous chemicals and as a constituent of motor fuels. It is a hydrocarbon that is likely to be found in tire fire emissions.

Sulfur dioxide, given the right conditions, turns into sulfuric acid. Sulfuric acid is used extensively in petroleum refining and the production of steel.

Vanadium is used as a catalyst in the oxidation of sulfur dioxides, oxides of nitrogen and other substances. In addition, vanadium is found in fuel oils.

Inhalation Reference Concentrations

Table 5 lists the chemicals with either a subchronic (SC) or chronic (C) inhalation reference concentration (RfC). Of the 14 chemicals with an RfC, eight exceeded one of the RfCs. It should be noted that the RfC is a value based upon a lifetime exposure, 24 hours/day. The values that were reported in these datasets were close to the tire fire. It is expected that some values detected close to the fire would exceed the RfC. A person should not be exposed to the emissions close to the tire fire because data indicates that prolonged and continual exposure to tire fire smoke can be harmful.

Top 25% of Chemicals

Table 6 lists the top 25% of chemicals from the Acurex and EPA data and their criteria for inclusion or exclusion as target chemicals. Thirty-three chemicals are included in this list. Of those 33, 14 had already been selected as target chemicals. The criteria for inclusion is given in Table 6. Of the remaining chemicals, four were eliminated because of low toxicity: bromochloromethane, indene, naphthalene, and trichlorofluoromethane. One of the chemicals was eliminated because the reported concentration was less than the reference concentration: ethyl benzene. Of the remaining 16, nine are a type of hydrocarbon and therefore, covered as coal tar pitch volatiles. Information could not be found on the remaining five, therefore, they were excluded.

Oil

Pyrolytic oil is the runoff that comes from melting tires and is also a concern from a tire fire. There is limited data on the chemical components of pyrolytic oil. Review of the data indicates that the primary constituents found in air due to a tire fire are the same components that are found in pyrolytic oil (Peterson et al, 1984; TAT, 1984). The major constituents in pyrolytic oil have been shown to be aliphatic, aromatic, and naphthenic hydrocarbons, specifically, benzo(a)pyrene, chrysene, benz(a)anthracene, oxides, carbon and nitrogen.

TABLE 5: CHEMICALS THAT HAVE SUBCHRONIC AND CHRONIC INHALATION REFERENCE CONCENTRATIONS BY MAXIMUM REPORTED VALUE AND REFERENCE CONCENTRATION

SOURCE	*CHEMICAL*	*Value (mg/m3)*	*SC RfC (mg/m3)*	*C RfC (mg/m3)*
Acu	**Barium**	**0.0035**	**0.005**	**0.0005**
EPA	**Chloride, ethyl**	**83**	**10**	**10**
EPA	**Cumene**	**0.094**	**0.09**	**0.009**
Acu	Cyclopentadiene	0.273	3	
EPA	Dichlorobenzene, 1,2-	0.0696	2	0.2
EPA	Dichlorobenzene, 1,4-	0.1187	0.7	0.7
EPA	**Dichloropropane, 1,2-**	**0.035**	**0.013**	**0.004**
EPA	Ethylbenzene	0.1554	1	1
Acu	Ethylbenzene	0.537	1	1
EPA	**Hexane**	**0.21**	**0.2**	**0.2**
EPA	Hexane, n-	0.158	0.2	0.2
Acu	Methylcyclohexene	0.03	3	3
EPA	Methylene chloride	0.0606	3	3
EPA	**Styrene**	**5.41**		**1**
Acu	Styrene	0.795		1
EPA	**Sulfuric acid**	**0.79**		**0.07**
EPA	**Toluene**	**6.7**	**2**	**0.4**

Bolded = values exceed Reference Concentration

TABLE 6: TOP 25% OF CHEMICALS BASED UPON MAXIMUM REPORTED VALUE. TARGET CHEMICALS IDENTIFIED

SOURCE	CHEMICAL	VALUE (mg/m3)	TARGET CHEMICAL Inclusion Criteria	TARGET CHEMICAL Exclusion Criteria
Acu	Acenaphthene	1.027	CA	
Acu	Acenaphthylene	0.897	CA	
Acu	Benzaldehyde	0.566		HC
EPA	Benzene	10.59	CA	
Acu	Benzene	3.872	CA	
Acu	Benzo(a)Pyrene	0.481	CA	
Acu	Benzo(g,h,i)Perylene	0.441		HC
EPA	Bromide	94.00		NI
EPA	Bromochloromethane	1.1360		LT/LC
EPA	Carbon Monoxide	100.00*	TLV	
EPA	Chloride	83.00	RfC	
EPA	Chloroform	2.0580	CA	
Acu	Chrysene	0.368	CA	
EPA	Coal Tar Pitch	4.2180	TLV/CA	
Acu	Dimethyl Benzene (Xylene)	0.789		LT/LC
Acu	Ethenyl Benzene	1.205		HC
Acu	Ethyl Benzene	0.537		RfC
Acu	Ethynyl Benzene	0.974		HC
Acu	Ethynyl,Methyl Benzene	1.347		HC
Acu	Fluoranthene	0.473		HC
EPA	Fluoride	32.00	TLV	
EPA	Hydrochloric Acid	4.000		NI
Acu	Indene	0.602		LT/LC
Acu	Isocyano Benzene	1.33		HC
Acu	Methyl Benzene (Toluene)	1.556	RfC	
Acu	Methyl Naphthalene	0.372		HC
Acu	Naphthalene	2.953		LT/LC
EPA	Naphthalene	1.3200		LT/LC
EPA	Nitrate-N	220.00		NI
EPA	Orthophosphate	280.00		NI
Acu	Phenol	0.473	CA	
EPA	Pyrylene	2.6230		HC
EPA	Styrene	5.4100	RfC/CA	
Acu	Styrene	0.795	CA	
EPA	Sulfate	230.00		NI
EPA	Toluene	6.700	RfC	
Acu	Trichlorofluoromethane	1.003		LT/LC
EPA	Xylene, M,P-	131.00		TLV
EPA	Xylene, O-	1564.00	TLV	

* = value is in ppm
TLV = Value Exceeds/Does Not Exceed 33% of Threshold Limit Value
CA = Chemical is Suspected or Confirmed Human Carcinogen
RfC = Value Exceeds/Does Not Exceed Subchronic and/or Chronic Inhalation Reference Con
HC = Hydrocarbon included as coal tar pitch volatiles
LT/LC = Low Toxicity/Low Concentration
NI = No Information

Ash

The residual material left after the fire is extinguished is termed "ash". A review of the literature yielded little data relating to ash. The Everett tire fire data indicated one of the major constituents of tire fire ash is zinc oxide. In fact, 12 heavy metals were identified of which zinc compounds accounted for 99% of all metals found. Data indicated that ash contained the same type of chemicals as the oil (e.g., hydrocarbons, metals, and various inorganics).

Exposure Assessment

The route of exposure, or an exposure pathway, is simply the pathway in which an individual is exposed to a chemical. There are four necessary elements to an exposure pathway: 1) a source and a mechanism of release for the chemical to enter the environment; 2) an environmental transport medium (e.g., air, ground water); 3) a location of potential human contact with the contaminated medium, known as the exposure point; and, 4) a human exposure route (e.g., drinking water, ingestion, inhalation, dermal absorption).

Chemical Release

The first step in assessing exposure risk is to determine possible chemical release sources and media. For a tire fire, the obvious source is the fire. The mechanisms of release include volatilization of gases and fumes, particulate generation, seepage of oil, and ash remaining after a fire.

Environmental Transport

The second step is to determine the transport medium. The primary transport media for the gases and particulates are volatilization and deposition into the air and onto the soil (via deposition). The transport mechanism for oil is runoff onto the soil, and into the ground water and surface water. The transport mechanism for ash is deposition onto the soil, with air transport representing a secondary transport mechanism. Soil, dust, and ash particulates can also be resuspended by wind.

Exposure Point

The third step is to identify possible human exposure points. There are two types of exposure points that need to be evaluated. The first is the area where an individual is exposed to the highest concentration of contaminants, e.g., a firefighter working in the smoke plume. The second exposure point is an area with a lower concentration of contaminants, but where a large or more sensitive population is exposed, e.g., area residents exposed to smoke, or contaminated soil or water.

Potential exposure points for air exposure would include populations of nearby communities, industrial or commercial areas, schools, nursing homes or any other points with human activity. Sources to obtain this information include: site vicinity surveys, topographic maps, aerial photos, county land-use maps and census data.

Points for soil and particulate exposure would occur in the area under the plume of smoke where particulates are deposited, as well as in the path of oil seepage.

The primary mechanism for chemical release to surface and ground water is seepage of oil and leaching of ash. The significant exposure points for surface water pathways depend upon downstream uses of the water. This would include both in-stream uses, such as water contact sports and fishing (resulting in consumption of contaminated fish) and withdrawal uses, such as domestic water supply, agricultural use and industrial use. Sources for identifying withdrawal points and uses include: site vicinity surveys, state water agency records, local water utility records, withdrawal permits, and the EPA Office of Drinking Water Federal Reporting Data System.

Exposure points from ground water include nearby wells in the direction of ground water flow. If modeling is not planned, determine the likely flow direction from geohydrologic data and assume that the closest domestic well in that direction is the highest individual exposure point. Locations and depths of public water supply wells should also be determined. In addition to domestic wells, locations of agricultural and industrial wells and any other relevant ground water uses must be determined.

Exposure Route

The fourth step is to determine the exposure route. For the air pathway the exposure route is inhalation or dermal absorption of particulates, fumes and gases.

The primary route of exposure for soil is through ingestion or dermal absorption of soil contaminated with oil, deposited particulates and ash. If site restrictions are not in place there is also a possibility of tracking contaminated soil to off-site locations. A possible indirect route of exposure from soil contamination is chemical uptake or deposition on plants with subsequent ingestion by humans.

Exposure routes for surface and ground water include ingestion of water and/or contaminated fish, inhalation of volatilized particulates from water, and dermal absorption from water contact sports and bathing in contaminated water. An indirect route of exposure would be watering crops that will uptake contaminated water and later be consumed by humans.

Exposure Pathways

In conclusion, the primary exposure pathway for a tire fire is the release of gases, fumes, particulates and ash into the air where workers and nearby residents inhale the particulates. Secondary exposure pathways include: 1) deposition of particulates and seepage of oil onto soil where either workers, nearby residents or possibly the family members of workers who track home the soil, ingest or inhale the soil particulates; 2) airborne particulates land on skin and are dermally absorbed; 3) a plant takes up the contamination via the soil and a human consumes the plant; and, 4) oil seeping into surface and ground water where individuals drinking the water, using the water downstream, or eating fish from the water are exposed (see Figure 2). The secondary pathways are less likely to occur and are dependent on the size of the fire and how long it has been burning.

Health Assessments

The exposure during a tire fire would be considered either an acute exposure that lasts for hours or a day, or a subchronic exposure that lasts a few days to weeks. It is very unlikely that a chronic exposure (years) would occur with a tire fire. Many of the toxicological actions of these chemicals are associated with chronic, long-term exposure, e.g., cancer. However, some of the target chemicals can have a serious health impact from an acute exposure (e.g., sulfuric acid, ethylene dichloride, and vanadium).

The target chemicals were grouped into three chemical categories: metals, hydrocarbons, (aromatic, polyaromatic, halogenated, and aliphatic) and inorganic gases and vapors. Table 7 lists the target chemicals by chemical category. This section will discuss the general health hazards associated with each of these categories. A brief discussion of the health hazards of each of the target chemicals can be found in Appendix A.

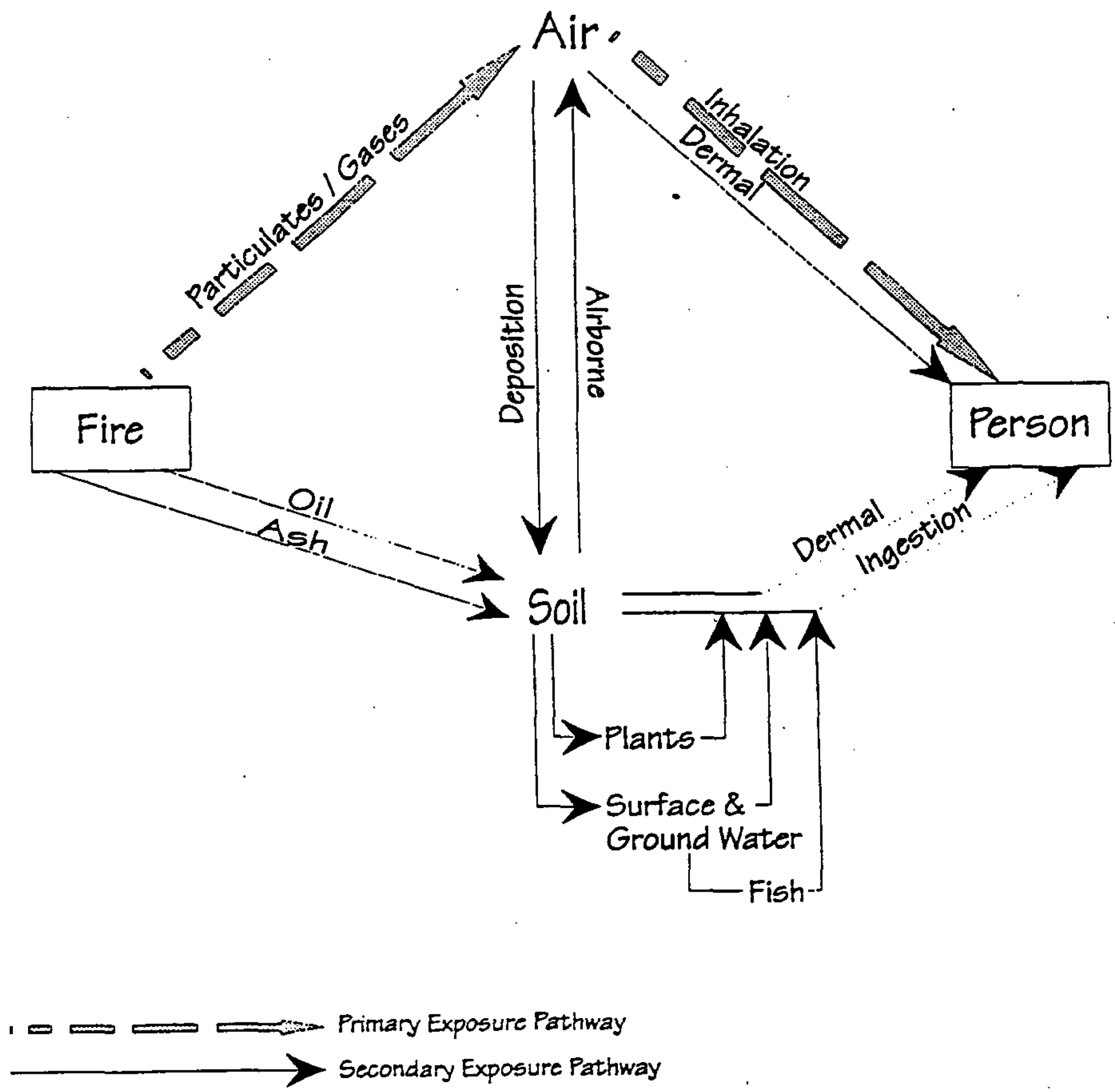
Hydrocarbons

These groups of chemicals share similar toxicological properties in that they generally are organic solvents. The most important toxicologic action of organic solvents is their capacity to affect the nervous system, particularly the functions of the brain. Such an affect is dependent upon the concentration and duration of exposure. Relatively low concentrations of aromatic hydrocarbons, such as benzene and toluene, can produce severe central nervous system (CNS) depression. The inhalation of vapors of organic solvents causes symptoms such as lightheadedness, giddiness and lack of coordination.

Solvents are capable of dissolving other organic substances such as fats and oils. Repeated exposure of solvents to the skin results in "defatting" of the skin. The skin becomes reddened, swollen, dry and cracked, a serious condition that allows other toxic chemicals to cross the skin barrier and enter the body.

Chemical pneumonitis is one of the more serious complications that can result from organic solvent poisonings; it occurs when solvents enter the lungs in liquid form. It is extremely unlikely that this condition would occur from exposure to emissions from a tire fire, but is mentioned due to its seriousness.

Figure 2: Potential Exposure Pathways associated with a Tire Fire



Adolfson Associates, Inc.
5309 Shilshole Ave. NW
Seattle, Wa 98107
(206) 789-9658

Field Sampling Data

Project ____________________

Well or Surface Site No. ____________________

Sample Designation ____________________

Date, Time ____________________

Weather ____________________

Hydrology Measurements:

(Nearest .01 ft.)	Date, Time	Method Used

Well Evacuation:

Gallons	Pore Volumes	Method Used	Date, Time

Sampling:

Sample	Date, Time	Method	Volume (ml)	Container Type	Depth Taken (feet)	Field Filtered (yes, no)	Preservative	Iced (yes, no)	Sampler Cleaning Method
									Non-Phosphatic detergent wash
									H2O rinse
									MeOH Rinse
									Distilled H2O rinse

Field Water Quality Tests:

Pore Volume Number	pH	Conductivity	Temp	Gallons

Notes:

Total No. of Bottles ____________ Signature: ____________________

ADOLFSON
ASSOCIATES, INC.
Environmental Analysis
*5309 Shilshole Ave. NW
Seattle, WA 98107
(206) 789-9658
FAX (206) 789-9684*

Chain of Custody/Laboratory Analysis Request

Date ______ Page ____ of ____

Project ______ No. ______
Client Contact ______
Address ______
Telephone ______
Sampler's Name ______ Phone ______
Sampler's Signature ______

Sample ID	Date	Time	Lab ID	Type	Analysis Requested																			No. of Containers
1																								
2																								
3																								
4																								
5																								
6																								
7																								
8																								

Relinquished by Adolfson Associates
Signature ______
Printed Name ______
Firm ______
Date/Time ______

Received by
Signature ______
Printed Name ______
Firm ______
Date/Ti ______

Relinquished by
Signature ______
Printed Name ______
Firm ______
Date/Time ______

Received by
Signature ______
Printed Name ______
Firm ______
Date/Time ______

Project Information
Shipping ID No. ______
VIA ______
Project ______

Sample Receipt
Total No. of Containers ______
Chain of Custody Seals ______
Received in Good Condition ______
Lab No. ______

Special Instructions/Comments

TABLE 7: TARGET CHEMICALS BY CHEMICAL CATEGORY

METALS:

- Arsenic
- Barium
- Chromium
- Lead
- Nickel
- Vanadium
- Zinc

INORGANIC GASES AND VAPORS:

- Carbon Monoxide
- Chloride
- Fluoride
- Sulfur Dioxide
- Sulfuric Acid

HYDROCARBONS:

Aromatic:

- Benzene
- Benzylchloride
- Cumene
- Phenol
- Styrene
- Toluene
- Xylene -O

Aliphatic

- Butadiene
- Hexane

Halogenated:

- Carbon Tetrachloride
- Chloroform
- Dichloropropane, 1,2-
- Ethylene Dichloride
- Hexachloroethane
- Methylene Chloride
- Trichloroethane, 1,1,2-
- Trichloroethylene

Polyaromatic:

- Acenaphthene
- Acenaphthylene
- Benz (a) anthracene
- Benzo (a) pyrene
- Benzo (b) Fluoranthene
- Chrysene
- Coal Tar Pitch
- Dibenz (a,h) anthracene

Metals

Metals released into the environment can be bioconcentrated and enter into the food chain. Most toxicologically important metals bind strongly to tissues and therefore, are excreted slowly. They tend to accumulate to a high degree with repeated exposure. The toxicological effects of metals cannot be generalized. Different metals have different tissue affinities and different toxicological actions. Specific discussions about metals can be found in Appendix A.

Inorganic Gases and Vapors

The inorganic gases and vapors include: carbon monoxide (chemical asphyxiant), fluoride and chloride (inorganic ions), sulfur dioxide, and sulfuric acid (irritants). These chemicals can be an acute exposure concern. Specific discussions about these chemicals can be found in Appendix A.

DISCUSSION

The target chemical selection process was aimed at determining a representative subset of chemicals that are released during a tire fire. The data used for this process consisted of field data from actual tire fire incidents, as well as data from a simulated tire fire in a laboratory setting. The list of chemicals used in the selection process was comprehensive (150 chemicals) and also appeared to be representative of tire fires when compared with other data from tire fires. The selection process resulted in 38 target chemicals based upon toxicity and concentration. They could be classified into three categories: hydrocarbons (primary category); heavy metals; and inorganic gases and vapors. With a few exceptions, the concentrations were relatively low. Those with higher concentrations (hydrocarbons, inorganic gases and vapors) are expected given the composition of tires.

The general health concern regarding these chemicals is that many of them are carcinogens. In addition, they are irritants to the mucous membranes, eyes and skin. An acute exposure to a high concentration could result in an adverse health effect. This type of exposure could occur to personnel working at the fire site. A chronic exposure, although unlikely, could also result in an adverse health effect.

The primary route of exposure is inhalation of particulates, gases and vapors. There are secondary exposure routes, such as dermal absorption, ingestion of soil particulates, exposed plants and contaminated water or fish. These secondary exposure routes are most significant in a situation where the fire burns for a long period of time and the runoff is not controlled.

Obviously, precautions should be taken to avoid inhalation of the plume within a close proximity to the fire. In addition, precautions should be taken to avoid dermal exposure to particulates and gases within close proximity to the fire. Exposure could result in CNS effects, dermatitis, and other adverse health effects. Personal Protective Equipment used to reduce and eliminate exposure is discussed in Section IV.

A set of chemicals has been identified through the selection process. Sampling for these target chemicals could provide information to assist with policy decisions regarding worker and community safety. Tire fire emissions are very toxic and could produce a significant hazard to workers at the fire site if proper protection is not taken. The risk to communities is expected to be much less; however this supposition should be confirmed with air monitoring data.

II. AIR MONITORING PLAN

INTRODUCTION

Exposure to air contaminants given off by a tire fire can be hazardous to the health of response personnel and nearby residents. The smoke emitted from smoldering tires contains hazardous substances, some of which are highly toxic and potentially carcinogenic (see Appendix A). Extra precautions and a high level of personal protection are warranted during the initial approach to a tire fire (refer to Section IV). Air monitoring can help protect the safety and health of response personnel, and provide information to assess the risk to nearby residents from exposure to harmful levels of airborne contaminants.

The purpose of air monitoring is to identify and quantify airborne contaminants. This information is used to assess the potential for exposure to harmful levels of air contaminants and determine the level of worker protection needed.

Two principal approaches are available for identifying and/or quantifying airborne contaminants:

- The on-site use of direct-reading instruments.

- Laboratory analysis of air samples obtained by gas sampling bag, collection media (i.e., filter, sorbent), and/or wet-contaminant collection methods.

Direct-reading instruments can be used to monitor for the presence of air contaminants during the initial approach and over the course of the tire fire to guide decisions regarding the location of the work zones and the level of protective equipment to be worn (see Section IV for discussion of work zone and protective equipment). In an emergency situation, such as the initial response to the fire, air sampling generally is limited to the use of direct-reading instruments. This equipment provides immediate data on emissions and allows response personnel to gain control of the situation quickly while minimizing the risk of exposure to hazardous substances.

For more complete information about air contaminants, the measurements obtained with direct-reading instruments must be supplemented by collecting and analyzing air samples. Air samples are collected for subsequent analysis to identify and quantify air contaminants.

OBJECTIVES

There are two primary objectives for air monitoring:

1) Assess safety and health risks

   Air monitoring is done to identify and quantify airborne contaminants to assess risks to the safety and health of response personnel and nearby residents. Continuous monitoring should be conducted using a direct-reading instrument to track changes in air contaminants that may occur during the course of the fire. Exposure levels of response personnel to target chemicals should not exceed the threshold limit values (TLVs).

2) Collect data for decisions regarding levels of protection and other protective measures

   The second objective is to confirm the decision regarding the establishment and location of work zones for various levels of protection. Direct-reading instruments can be used to monitor for the presence of air contaminants to help establish the location of decontamination and work zones. Once the concentrations and distribution of airborne contaminants are understood, appropriate adjustments in the level of protective equipment can be made. Working in high level protective equipment such as self-contained breathing apparatus (SCBA) and fully encapsulated suits is arduous, time consuming, and poses its own safety risks. However, it may be possible to reduce the level of protection required in designated areas. For instance, if monitoring results show concentrations consistently below the Immediately Dangerous to Life and Health (IDLH) points, and the dynamics of the situation are clearly understood, it may be appropriate to down-stage from SCBA to an air purifying respirator in certain areas, or to discontinue the use of respirators. It is important to verify that personal protective equipment is not being removed in contaminated areas. (See Section IV).

EQUIPMENT

<u>Direct-Reading Instruments</u>

Of significant importance, especially during initial entries, is the potential for IDLH conditions or oxygen deficient atmospheres. Direct-reading instruments can be useful in identifying IDLH conditions, toxic levels of airborne contaminants, and flammable atmospheres. Direct-reading instruments with three sensors can monitor flammable vapors, oxygen content and carbon monoxide concentrations simultaneously. These three-way meters would be the best type of meter to use during a tire fire because there is concern about flammable vapors, oxygen content and carbon monoxide exposure. Table 8, excerpted from Occupational Safety and Health Guidelines for Hazardous Waste Site Activities, provides an overview of available monitoring instrumentation and their specific operating parameters. Periodic monitoring of conditions is critical, especially if exposures may have increased since initial monitoring or if new site activities have commenced.

TABLE 8: DIRECT-READING INSTRUMENTS FOR GENERAL SURVEY

Combustible gas indicator (CGI)	
Hazard Monitored:	Combustible gases and vapors.
Application:	Measures the concentration of a combustible gas or vapor
Detection Method:	A filament, usually made of platinum, is heated by burning the combustible gas or vapor. The increase in heat is measured. Gases and vapors are ionized in a flame. A current is produced in proportion to the number of carbon atoms present.
General Care/Maintenance:	Recharge or replace battery. Calibrate immediately before use.
Operating Time:	Can be used for as long as the battery lasts, or for the recommended interval between calibrations, whichever is less.
Flame Ionization Detector (FID) with Gas Chromatography Option.	
Example:	Foxboro OVA.
Hazard Monitored:	Many organic gases and vapors.
Application:	In survey mode, detects the concentration of many organic gases and vapors. In gas chromatography (GC) mode identifies and measures specific compounds. In survey mode, all the organic compounds are ionized and detected at the same time. In GC mode, volatile species are separated.
General Care/Maintenance:	Recharge or replace battery. Monitor fuel and/or combustion air supply gauges. Perform routine maintenance as described in the manual. Check for leaks.
Operating Time:	8 hours; 3 hours with strip chart recorder.
Portable Infrared (IR) Spectrophotometer	
Hazard Monitored:	Many gases and vapors.
Application:	Measures concentration of many gases and vapors in air. Designed to quantify one or two component mixtures.
Detection Method:	Passes different frequencies of IR through the sample. The frequencies absorbed are specific for each compound.
Care/Maintenance:	As specified by manufacturer.

Air Sampling Methods

If the release of air contaminants continues long enough, air sampling equipment can be brought in to determine the type and quantity of airborne contaminants present during the fire. To assess air contaminants, air sampling pumps with the appropriate collection media should be placed at sampling locations as described in the next section. Air sampling methods for the tire fire target chemicals are listed in Table 9.

SAMPLING PROCEDURES

Sample Locations

The preparation of a site map is an essential task when sampling an area where existing maps are not available. The map is used to indicate the location of sample points and should contain information such as compass direction and environmental conditions.

Establish sampling sites in the following locations (mark sample sites on site work map):

1. Upwind - Establish background air contaminant levels.

2. Support Zone - Ensure that command post and other facilities are located in a "clean" area.

3. Contamination Reduction Zone - Ensure that decontamination workers are properly protected and that on-site workers do not remove protective gear in a contaminated area.

4. Exclusion Zone - Verify, continually confirm, and document selection of proper levels of protection as well as provide continual records of air contaminants.

5. Downwind - Indicate if any air contaminants are leaving the site.

Sampling locations should be located upwind to check background levels, immediately downwind to identify contaminants, and at the site perimeter or fence line to determine if measurable concentration of air contaminants are leaving the site. If there are indications of airborne hazards in populated areas, additional sampling stations should be placed downwind.

Use direct-reading instruments or detector tubes to monitor each sample site and record the time and readings using a site log.

Collect air samples at each site using the appropriate time periods and method to quantify specific air contaminants (see Table 9).

TABLE 9: AIR SAMPLING METHODS FOR TIRE FIRE TARGET COMPOUNDS

TARGET COMPOUND	COLLECTION MEDIA	COLLECTION METHOD	ANALYTICAL METHOD
Polyaromatic Hydrocarbons PAH's	Washed XAD-2, 37 mm PTFE	NIOSH 5515	GC-PID
Hydrocarbons BP 36-126 C	Charcoal	NIOSH 1500 NIOSH 1501 NIOSH 1003	GC-FID GC/MS
Volatile Organic Compounds	Tenax(R)	EPA TO-1 EPA TO-2	GC-MS
Metals	37 mm 0.8 um MCEF filter	NIOSH 7300	ICP-AES
Total Suspended Particulate	Glass Fiber Filter	EPA	GR
Inorganic Acids	Washed Silica Gel	NIOSH 7903	IC

LEGEND:
GC-PID = Gas Chromatography - Photoionization Detector
GC-FID = Gas Chromotography-Flame Ionization Detector
GC/MS = Gas Chromotography/Mass Spectrometry
ICP-AES = Inductively Coupled Argon Plasma Atomic Emission Spectroscopy
IC = Ion Chromotography
GR = Gravimetric

Sampling Frequency

Sampling should be conducted initially and when there is reason to believe exposure to air concentrations may have risen to levels above the permissible exposure limits. Meteorological conditions must be considered. Data on wind speed and direction, temperature, humidity, and barometric pressure should be obtained and recorded. It may be necessary to establish a site weather monitoring station to assist in decision-making. In addition, weather information is needed for calibrating instruments, dispersion modeling, and determining populations at risk from exposure to air contaminants.

Field Reference Sheet

When selecting field sampling equipment and designing a sampling plan, consideration must be made of: toxicity and concentration of compounds suspected; which if any compounds occur in a consistent relationship to others; and sampling equipment availability. A sample field reference sheet for a tire fire is shown in Appendix D. Appendix D shows a simplified field guide with a short list of target chemicals, sampling equipment and, where available, action levels. A field reference sheet should be designed prior to a tire fire using the list of target chemicals (Tables 2 and 7), the list of instruments for a general survey (Table 8), the list of air sampling methods (Table 9), and the list of available equipment.

DOCUMENTATION

All monitoring results must be documented and kept on file at the site. The following documentation should be maintained:

1) AIR MONITORING PLAN - A general description of the air monitoring conducted, the instruments and equipment used, the location of sample sites and frequency of sample collection.

2) SAMPLE LOG - A list of the date and time samples were collected in chronological order for each sample site.

3) SAMPLE SITE MAP - A site map on which the location of the sampling sites relative to the source is marked.

4) FIELD SAMPLE DATA - A data sheet on which the sample descriptions, field sampling data, and the results of direct-reading instruments are recorded.

5) INSTRUMENT CALIBRATION DATA - Calibration check sheets with instrument reference number, date and initials.

6) INSTRUMENTS USED - A listing should be kept of the type of instrument, model number, serial number, and owner/organization of the instrument.

ACTION LEVELS

Since decisions regarding the need for precautions to protect the safety and health of response personnel and nearby residents are based in part on air monitoring data, a high level of confidence in the accuracy of the data is required. An understanding of the limitations of air monitoring results is critical, along with experience and good judgment, to properly interpret the direct-reading instruments and the air sample results.

The TLVs for the target compounds are listed in Table 10. The chemicals for which there are no TLVs are PAHs and are included under coal tar pitch volatiles. Table 11 lists the action levels pertaining to 1) explosive atmospheres; 2) oxygen content; 3) organic gases and vapors; 4) inorganic gases and vapors; and 5) particulates. For organic and inorganic gases and vapors and particulates, the action level is dependent upon the contaminant type. Oxygen should be monitored to be sure the combustible gas indicator is operating correctly. At low oxygen levels the combustible gas indicator does not function properly. In addition, if oxygen-deficient atmospheres are detected, the use of level A personal protection (SCBA) is mandated.

Caution must be exercised when using the action level guidelines in Table 11 for interpreting air monitoring instruments. The limitations of the instruments, the meteorological conditions, and the proximity of the surrounding communities are examples of considerations that must be taken into account when interpreting results.

DISCUSSION

The air contaminants from a tire fire contain many substances that could be hazardous to on-site personnel and nearby residents. Air monitoring should be conducted in order to assure proper protective measures are being taken and to assess the safety and health risks.

Air monitoring can be conducted either with direct-reading instruments or by collecting and analyzing air samples. The direct-reading instruments give instant readings and are helpful in determining if explosive or oxygen deficient conditions exist. The air samples assist in determining and quantifying the chemicals that are being released from the fire.

The action levels pertaining to explosive and oxygen deficient conditions are listed in Table 11. The action level for organic and inorganic gases and vapors and particulates are dependent upon the specific chemical. In general, samples at or above 33% of the TLV indicate continued air monitoring should be conducted to determine the extent of downwind contamination and if further action is needed.

It should also be noted that a tire fire can burn or smolder for months. The smoldering phase is not a hot burning phase and does not result in complete combustion; therefore, this phase can have excessive emissions. Air sampling should continue throughout all phases of the fire including the smoldering phase to assure that proper health and safety decisions can be made.

TABLE 10: TARGET CHEMICALS AND THEIR RESPECTIVE THRESHOLD LIMIT VALUE

Chemical	TLV (mg/m3)
Acenaphthene	NA
Acenaphthylene	NA
Arsenic	0.2
Benz(a)Anthracene	NA
Benzene	32
Benzo(a)Pyrene	NA
Benzo(b)Fluoranthene	NA
Benzylchloride	5.2
Butadiene	22
Carbon Monoxide	25
Carbon Tetrachloride	5
Chloride	NA
Chloroform	49
Chromium	0.05
Chrysene	NA
Coal Tar Pitch	0.2
Cumene	246
Dibenz(a,h,)Anthracene	NA
Dichloropropane, 1,2-	347
Ethylene Dichloride	40
Fluoride	2.5
Hexachloroethane	9.7
Hexane	176
Lead	0.15
Methylene Chloride	50
Nickel	1
Phenol	19
Styrene	213
Sulfur Dioxide	2
Sulfuric Acid	1
Toluene	188
Trichloroethane, 1,1,2-	10
Trichloroethylene	50
Vanadium	0.05
Xylene, O-	434
Zinc	0.01

NA = Not Available

TABLE 11: ACTION LEVELS

Explosive atmosphere:	
Action Level	Action
<10% LEL	Continue investigation.
10%-25% LEL	Continue on-site monitoring with extreme caution as higher levels are encountered.
>25% LEL	Explosion hazard. Withdraw from area immediately.
Oxygen Content:	
Action Level	Action
<19.5% O_2	Monitor wearing self-contained breathing apparatus. NOTE: Combustible gas readings are not valid in atmospheres with <19.5% oxygen.
19.5%-25% O_2	Continue investigation with caution. Deviation from normal level may be due to presence of other substances.
>25% O_2	Fire hazard potential. Discontinue investigation. Consult a fire safety specialist.
Organic gases and vapors:	
Action Level	Action
Depends on contaminant	Consult standard reference manuals for air concentration/toxicity data. Action level depends on PEL/TLV. Action Level is 1/3 the current standard. See Table 10.
Oxides of Carbon, Nitrogen and Sulfur:	
Action Level	Action
Depends on contaminant	Consult standard reference manuals for air concentration/toxicity data. Action level depends on PEL/TLV. Action Level is 1/3 the current standard. See Table 10.
Particulates:	
Action Level	Action
Depends on contaminant	Consult standard reference manuals for air concentration/toxicity data. Action level depends on PEL/TLV. Action Level is 1/3 the current standard. See Table 10.

LEL = Lower Explosive Limit
PEL = Permissible Exposure Limit
TLV = Threshold Limit Value

III. GENERAL SURFACE AND GROUND WATER SAMPLING PLAN

INTRODUCTION

This section includes the general sampling protocols and quality control procedures for surface and ground water sample collection at a tire fire site. Surface and ground water sampling may be desired to determine the extent of contamination emanating from a tire fire. A list of target sample parameters and corresponding EPA sample method numbers was determined along with a list of necessary sample collection equipment. General ground water monitoring well installation procedures were also included because it was assumed that at most locations monitoring wells would not be in place prior to a tire fire.

APPROACH

A literature search was conducted to obtain data from previously reported tire fires with water quality information. Only four fires reported information relating to water quality; these fires occurred in Duryea, Pennsylvania (1991), Everett, Washington (1984), Winchester, Virginia (1983), and Beith in Ayrshire, Strathclyde, Great Britain (1978). These reports were reviewed and a list of target chemicals was derived from the chemicals detected from these fires. This list includes the chemicals that are most likely to be found in surface or ground water after a tire fire. The list is similar to the list of target chemicals from Section I with the addition of several hydrocarbons. Sampling frequency recommendations were also determined based upon practices conducted at other fire sites. It should be noted that the tire fire in Everett, Washington occurred at an existing landfill. Some of the chemicals detected in the ground water at the Everett site may be related to the landfill and not to the tire fire, however the chemicals detected were similar to those detected at the other fires.

The Goodyear Tire and Rubber Company Chemical Material Safety Department was contacted and they provided a list of decomposition products of burning tires. The list of chemicals Goodyear noted corresponds with the list developed from the reports listed above and confirms the finding from the target chemical selection process (Section I).

FINDINGS

Target Chemicals

A list of target chemicals released from a tire fire that are likely to be detected in surface and/or ground water is included in Table 12. This list contains the chemicals recommended for monitoring. Corresponding EPA sample method numbers are also included in Table 12.

TABLE 12. LIST OF RECOMMENDED SAMPLE PARAMETERS

Recommended Surface and Ground Water Sample Parameters

Chemical	EPA Method Number	Chemical	EPA Method Number
Acenaphthylene	8270	Methyl-O-methylcaprolactam, N-	8270
Acenaphthene	8270	Methylbenzothiazole, 2-	8270
Acetophenone	8270	Methylcaprolactam, N-	8270
Anthracene	8270	Methylnaphthalene, 2-	8270
Azulene	8270	Naphthalene	8270
Benzene	8020	Nitrophenol, 2-	8270
Benzo(a)anthracene	8270	Nitrosodiphenylamine, N-	8070
Benzo(a)pyrene	8270	Phenanthrene	8270
Benzo(b)fluoranthene	8270	Phenol	8040
Benzo(g,h,i)perylene	8270	Pyrene	8270
Benzoic acid	8270	Styrene	8020
Benzonitrile	8270	Toluene	8020
Benzothiazole	8270	Toluic acid, 3-	8270
Bromoquinoline, 3-	8270	Tolunitrile, p-	8270
C_3-Benzamide	8270	Xylene, m/p-	8020
Caprolactam	8270	Xylene, o-	8020
Chrysene	8270	Total Organic Carbon*	415.1
Cyanide	335.2	Arsenic	7060
Cyanobenzoic acid, 3-	8270	Beryllium	6010
Dimethylbenzamide, N,N-	8270	Cadmium	6010
Dimethylphenol, 2,4-	8270	Chromium	6010
Ethanol, 2-(2-n-Butoxyethoxy)	8015 modified	Copper	6010
Ethylbenzene	8020	Lead	6010
Fluoranthene	8270	Nickel	6010
Fluorene	8270	Sulfate	300.0
Indeno (1,2,3-cd)pyrene	8270	Sulfite	377.1
Isophorone	8270	Zinc	6010

Recommended Soil Sample Parameters

Chemical	EPA Method Number	Chemical	EPA Method Number
Polynuclear Aromatic Hydrocarbons*	8270	Cyanide	9010
Arsenic	7060	Lead	6010
Beryllium	6010	Nickel	6010
Cadmium	6010	Sulfate	300.0 Mod.
Chromium	6010	Zinc	6010
Copper	6010		

* These analyses are useful for initial screening.

NOTE: Laboratory standards may not be available for all of these compounds in the event of emergency sampling. However, if sampling is routine, most laboratories can generate standards for the less common compounds.

General Surface Water Sampling Procedures

Surface Water Sample Locations and Frequency. Sample collection locations should be selected to provide a representation of the water and oil flowing from the tire fire site. At least one surface water sample should be collected from each direction of oil and surface water flow, including fire fighting efforts, and at least one "upstream" control location. More sample locations may be necessary if the tire fire site is adjacent to a water body. Nearby surface water bodies (e.g., wetlands, creeks, lakes, streams) should be monitored immediately for the target chemicals listed in Table 12. It should be noted that the direction of surface water flow may change with fire fighting efforts. Sampling should be repeated if the flow of runoff changes due to fire fighting efforts. Drainage ditches can be sampled by creating a ponded area from which sample containers can be filled as described below. Sample collection locations in flowing water are preferable to ponded water if the option exists. Ponded water may be subject to question due to settling. Ponded water locations are recommended for sediment sample collection (refer to the Soil Samples Discussion below).

Depending upon the size of the tire fire and the degree of contamination, sampling may be necessary on a bi-weekly basis. It is recommended that samples be collected during both storm and non-storm events if possible. Samples should be collected on a bi-weekly basis at least until the extent of contamination can be ascertained. A less frequent schedule may be developed after the concentrations of compounds decline or are not present, or it is determined that a health risk is not present. Receiving waters should also be monitored for the presence of contaminants. It may be that contaminants are found in surface water even after the fire is extinguished due to continued run-off.

Depending upon the location of the tire fire in relation to surface water bodies, biological sampling may be appropriate. The use of live boxes may be beneficial in determining the effects of contamination on aquatic organisms in adjacent receiving waters. Indigenous species should be utilized if live boxes are implemented. Sampling methods should be consistent with accepted EPA protocols.

Surface Water Sample Collection. Surface water samples may be collected by grab sample methodology. Sample containers may be placed in the flow or ponded water such that the water is allowed to flow into the container with minimal agitation. Care must be taken not to spill any preservatives that may be in the sample container. Disposable gloves should be worn during sample collection, and changed between collection locations. A list of general sample equipment is listed in Table 13.

The general surface water sampling procedure is as follows:

- Measure water depth in ponded area or stream.
- Measure and record field parameters of pH, conductivity and temperature.
- Label sample bottles with indelible ink.
- Transfer surface water to sample containers with minimal agitation for organic sample analysis.

TABLE 13: SAMPLING EQUIPMENT LIST

Surface Water Sampling Equipment	• pH/conductivity/temperature meter • boots/waders • disposable vinyl gloves • tape measure
Ground Water Sampling Equipment	• electric water level indicator • tape measure • Teflon bailer (double check-valves for organic sample collection) • nylon rope • peristaltic pump • power source (battery or generator) • Tygon or silicon tubing • pH/conductivity/temperature meter (same as above) • disposable vinyl gloves
Soil/Sediment Sampling Equipment	• stainless steel or Teflon spoons • stainless steel bowl • aluminum foil
Decontamination Equipment (not necessary for surface water grab samples)	• non-phosphatic detergent (e.g., Liquinox, Alconox) • methanol • nitric acid • wash tub • brushes • distilled water
Miscellaneous Equipment	• field data sheets/field notebook • Chain-of Custody/Laboratory Analysis Request forms • indelible markers • pH and conductivity calibration standards • camera

- Document activities including weather, sample identification, date and time, etc.
- Ship samples to the analytical laboratory within 24 hours of collection.
- Follow chain-of-custody procedures as outlined below.

Sample Container Preparation and Preservatives. All sample containers will be prepared and provided by the analytical laboratory. Samples will be preserved as per recommendations given in Methods for Chemical Analysis of Water and Wastes, EPA-600/4-79-020 March 1979. The type and size of container used for each parameter, as well as any preservative utilized, should be recorded on the field sampling data sheets.

Field Measurements. Time-sensitive parameters, i.e., temperature, pH and specific conductance, should be measured in the field at the time of sample collection. Field instruments should be calibrated using known standard solutions prior to sample collection. Conductivity, pH and temperature measurements may be obtained using a portable Corning meter or similar instrument. Instrument maintenance should be performed as necessary by the manufacturer.

General Ground Water Sampling Procedures

Ground water sampling is recommended in the event of extensive surface water and soil contamination. Due to the highly permeable nature of the soils in much of Pierce County, contaminants may move through the subsurface rapidly. Ground water sampling may not be necessary, however, if contaminated soil is removed quickly.

General ground water sampling and monitoring well installation procedures are described below. Site specific investigation will be necessary prior to the installation of monitoring wells. Ground water flow direction should be determined to facilitate monitoring well locations. A hydrogeologist should be consulted prior to and during monitoring well installation to assess the hydrogeologic site conditions. Other factors to be considered include soil conditions at the site, terrain, and location of nearby domestic well water.

Drilling and Borehole Logging. Drilling for the installation of ground water monitoring wells may be performed using hollow stem auger or other suitable drilling methods. All monitoring well installations and abandonments should be performed by a licensed driller. Drilling equipment should be steam cleaned prior to drilling the first boring, and between borings.

All drilling activities should be supervised by a hydrogeologist. A continuous log of subsurface soils should be prepared for each boring location. Each boring log should include the following information:

- Name and location of project;
- Boring and/or well number;
- Drilling contractor and drilling method;
- Detailed description of soil samples and drill cuttings including designation according

to the Unified Soil Classification System (USCS);
- Start and finish of drilling;
- Date and time of water level measurements.

Monitoring Well Installation. Upon the completion of each boring, a single completion monitoring well will be installed. Each monitoring well may consist of Schedule 40, 2-inch diameter flush threaded PVC well screen with riser pipe. All screen will be factory slotted with 0.010-inch size slots. If appropriate, an annular sand pack will be placed around and extended approximately three feet above the screened interval, and placed above the natural heave (if present). A bentonite seal will be placed from three feet above the screen to ground surface. After the completion of each well, the screened zone will be developed using pumping and/or surging techniques. Completed wells will be secured with locking caps, steel security casings, and traffic posts as necessary to prevent damage and/or vandalism.

Well Development. Following the installation of each monitoring well, and prior to sampling, the well's screened zone should be developed by pumping, bailing, and/or surging until the discharge water is free of sediment and non-turbid, or field measurements of pH, temperature and conductivity have stabilized or show no further change.

Ground Water Measurements. Depth to water measurements may be obtained with an electric water level indicator (ACTAT Olympic well probe or similar instrument) for all ground water sample locations. Water levels should be measured to the nearest 0.01 foot prior to sample collection. The indicator probe should be rinsed with distilled water prior to use in each well. All measurements should be taken from a marked point on top of the PVC monitoring well casing. Each measurement should be recorded on field sampling data sheets and include the date and time of measurement.

Stagnant water may be purged from well casings prior to ground water sampling by either bailing or pumping the well. Either method is acceptable for well evacuation; pumping is less labor intensive. Bailing should be performed by using a Teflon bailer lowered into the well using nylon rope. Pumping may be performed using a peristaltic pump with Tygon or Teflon tubing lowered into the well. Care should be taken not to introduce contaminants into the wells while purging.

The general ground water sampling procedure is as follows: (refer to Table 13 for a list of necessary sampling equipment)

- Measure static water level with an electric water level indicator.
- Purge a minimum of three well casing volumes or until pH and conductivity stabilizes. Well casing volumes are calculated as follows: $V = 3.14 \times R^2 \times L \times 7.48$ gallons/cubic foot, where V = volume (gallons), R = monitoring well radius (feet) and L = length of static water in well (feet).
- Measure and record field parameters of pH, conductivity and temperature.
- Label sample bottles with indelible ink.
- Transfer ground water to sample containers with minimal agitation for organic

sample analysis.

- Document activities including weather, sample identification, date and time, etc.
- Ship samples to the analytical laboratory within 24 hours of collection.
- Follow chain-of-custody procedures as outlined below.

Soil Samples

The collection of soil samples from areas of run-off or nearby stream beds may be useful in determining the extent of contamination, the contaminants that may enter the ground water system, and aid in clean-up and disposal of the soil as necessary. Composite sampling may provide more representative information regarding the extent of soil contamination and provide a cost savings. Based on review of data from other tire fires, recommended parameters for analysis are included in Table 12.

Sample collection should be conducted in areas of visual contamination from oil or water run-off, deposition, habitat areas, and/or areas with sensitive resources. Areas of ponded water are also recommended for soil collection, as these areas tend to collect pollutants that settle out of the water. Soil collection should be conducted utilizing stainless steel or Teflon spoons (refer to Table 13 for list of necessary sampling equipment). Soil should be placed in containers with minimal agitation for volatile organic analysis. Composite samples should consist of equal amounts of soil from each location. Combined individual samples should be thoroughly homogenized in a stainless steel bowl to provide one sample for analysis. Samples for volatile organics should be composited in the laboratory. Decontamination procedures should be followed as described below.

Sample Labeling, Shipping and Chain-of-Custody

Sample Labeling. Sample containers should be labeled prior to sample collection, and include the following information:

- Sample number
- Name of collector
- Date and time of collection
- Place of collection
- Sample preservation (if appropriate)

Sample Shipping. Samples should be shipped or delivered to the analytical laboratory using the following procedure:

- Sample containers should be transported in a sealed ice chest or other suitable container. Samples should be maintained at 4° C until receipt at the laboratory. The ice chest drainage hole should be secured in the event of sample container leakage.
- Glass sample containers should be separated in the shipping container by absorbent

A solvent, such as hexane, may be necessary to remove pyrolysis oil from the sampling equipment. Care must be taken if the solvent is highly flammable. The solvent rinse, followed by distilled water rinse, would precede the methanol rinse (step 3) in the decontamination sequence listed above. Minimal quantities of solvents should be used for decontamination. Care should be taken to ensure that uncontrolled releases of large quantities of solvents do not occur. Decontamination solvents should be contained and disposed of properly off-site.

Quality Control

Quality Control Samples. Quality control samples, including field blanks, transport blanks and duplicate samples, should be included in the sampling protocol.

Duplicate ground water samples can be obtained by alternately filling like sample containers for two sample sets until all containers are full. Approximately five percent of ground water samples should be collected as duplicates. Surface water sample duplicates may be unobtainable due to lack of sufficient water volume for duplicate sample collection (depending upon the sample location). Alternate locations may be selected for duplicate sample collection, if some sites are unsuitable.

Field blanks (method blanks) should be taken for each separate sample bottle type and analyzed in the laboratory with the other samples. Field blank samples can be obtained following equipment decontamination by collecting distilled water that has passed through the sampling equipment. Field blanks should be collected at a rate of approximately one in every 20 samples.

Transport blanks, if requested, will be provided by the analytical laboratory, accompany the shipment of sample bottles to the site, and will return to the laboratory for analysis with the sample shipment. Each type of sample bottle will be filled with deionized water at the laboratory, accompany sample shipment, and be returned to the laboratory, unopened, for analysis. Transport blanks may be analyzed at a rate of approximately one set per sampling event.

Documentation

Accurate documentation of field activities should be maintained using field logs, field data forms, correspondence records and photographic slides. Entries should be made in sufficient detail to provide accurate record of field activities without reliance on memory.

DISCUSSION

Sampling of ground water, surface water, and soil may not be necessary at all tire fires. It is dependent upon factors such as the size of the fire, the proximity of the fire to surface waters, the degree of run-off water and oil, the proximity of domestic water wells to the fire, and soil material to prevent breakage.

- Ice should be placed in separate sealed plastic bags, or utilize "blue" ice.
- All sample shipments must be accompanied by a chain-of-custody laboratory analysis request forms. A plastic bag containing the chain-of-custody forms should be taped to the inside lid of the ice chest. The ice chest should be sealed with tape.
- The shipper's office, name, and address should be placed on the ice chest.
- Samples should be shipped (or delivered) to the analytical laboratory within 24 hours of sample collection.

Chain-of-Custody. Upon transfer of sample possession to subsequent custodians, a chain-of-custody form should be signed by the persons transferring custody of the sample container. Upon receipt of the samples at the laboratory, the shipping container seal will be broken and the condition of the samples will be recorded by the receiver. Chain-of-custody records will be included in the analytical report prepared by the laboratory.

Field sampling data/chain-of-custody forms should be used during sampling. Examples of these forms are included in Appendix B. These sheets provide documentation of the following information:

- Project name
- Location and sampling source
- Time and date of sampling
- Pertinent hydrologic data, e.g., depth to water
- Sampling method, e.g., bailer
- Sample preservation, if any
- Type of sample, volume and type of container
- Weather
- Field measured parameters, e.g., pH, temperature, specific conductance
- Samplers name
- Sample storage
- Comments, e.g., appearance of sample
- Total number of bottles
- Type of analyses required

Decontamination Procedures

All non-dedicated sampling equipment should be thoroughly decontaminated prior to sample collection at each location. The following sequence of equipment decontamination should be followed prior to each sampling station:

1. Non-phosphatic detergent wash (e.g., Liquinox or Alconox)
2. Distilled water rinse
3. Reagent-grade methanol rinse
4. Distilled water rinse

type and ground water depth. The need and frequency of surface water, ground water and soil samples should be assessed on a case-by-case basis. Other studies indicated that oil and water run-off from a tire fire can contain elevated levels of target compounds and contaminate surface waters. All precautions should be taken to avoid run-off entering nearby surface waters or nearby dry wells that discharge to the subsurface.

IV. PERSONAL PROTECTION AND SAFETY

INTRODUCTION

If not protected properly, personnel responding to a tire fire could be exposed to hazardous substances, including, metals, PAHs, carbon monoxide and acids, as well as excessive heat (see Section I). Possible adverse health effects include cancer, CNS effects, irritation and damage to the eyes, nose, and respiratory tract, dermatitis and heat exhaustion (see Appendix A). Clearly, personnel responding to a tire fire should be well trained and protected to assure their health and safety.

In order for a chemical to have an adverse effect it must first gain entry into the body. The skin, eyes and the respiratory tract are usually the first tissues exposed to a chemical. These tissues provide some protection for the internal organs; however, they are often damaged or the chemical permeates through them. Personal protective equipment is used to eliminate exposure to the skin, eyes and respiratory tract so that they are not damaged and the internal organs are protected.

The selection of proper personal protective equipment is essential to assuring the protection of workers. This section gives the criteria that should be applied when selecting protective equipment for personnel responding to a tire fire. In addition, safety and general personal decontamination procedures are discussed. This section is not meant to take the place of an emergency response plan. The intent is to give an overview of the issues that should be considered when developing a response plan for a tire fire.

PERSONAL PROTECTIVE EQUIPMENT

The choice of personal protective equipment (PPE) is dependent upon what hazards are found on site. When assessing the hazards, consideration should be given to all potential chemical (toxins), physical (heat) and environmental hazards (atmospheric conditions, terrain). In the case of a tire fire, the potential chemical and physical hazards are fairly well understood, however, the environmental hazards are site specific.

Chemical Protective Clothing

The most important factors in determining what type of protective clothing to choose are, what chemicals are present and in what concentration. In the case of a tire fire, the chemicals and concentrations that would be on site are generally understood to be hydrocarbons, metals, and inorganic gases such as sulfur dioxide and carbon monoxide. It should be noted that if a fire occurs where there are other products burning in addition to tires, air monitoring should be conducted to assure there are not additional chemicals present that a worker needs to be protected against (see Section II).

There are four generally accepted levels of PPE, based upon respiratory protection and skin protection. The levels are A, B, C and D, in which level A offers the highest level of protection. Table 14 describes these four levels of protection, and the recommended equipment and optional equipment for each.

In selecting the type of protective clothing that should be worn, it should be understood that there is no single protective material that protects the wearer from all chemicals. All materials will decompose, be permeated or otherwise fail to protect under given circumstances. Manufacturers rate their protective clothing for degradation and permeation rates and breakthrough times. When using the rating tables, keep in mind that the ratings are based upon exposure to individual products and chemicals, while in the case of a tire fire there are multiple chemicals. Decisions regarding type of material should be made by experienced personnel in conjunction with the suit manufacturers. *It should be noted that fire fighting bunker gear is not considered an EPA protection level standard. Bunker gear is the best available equipment to protect against heat, however, it provides only limited protection against chemicals. An on-site decision would have to be made regarding the wearing of bunker gear versus chemical resistant clothing.* If a is fire drafting (providing a strong air current toward the fire at the ground level), it is possible that bunker gear would provide the necessary level of protection. However, this decision would need to be made by a trained emergency response commander on a case by case basis.

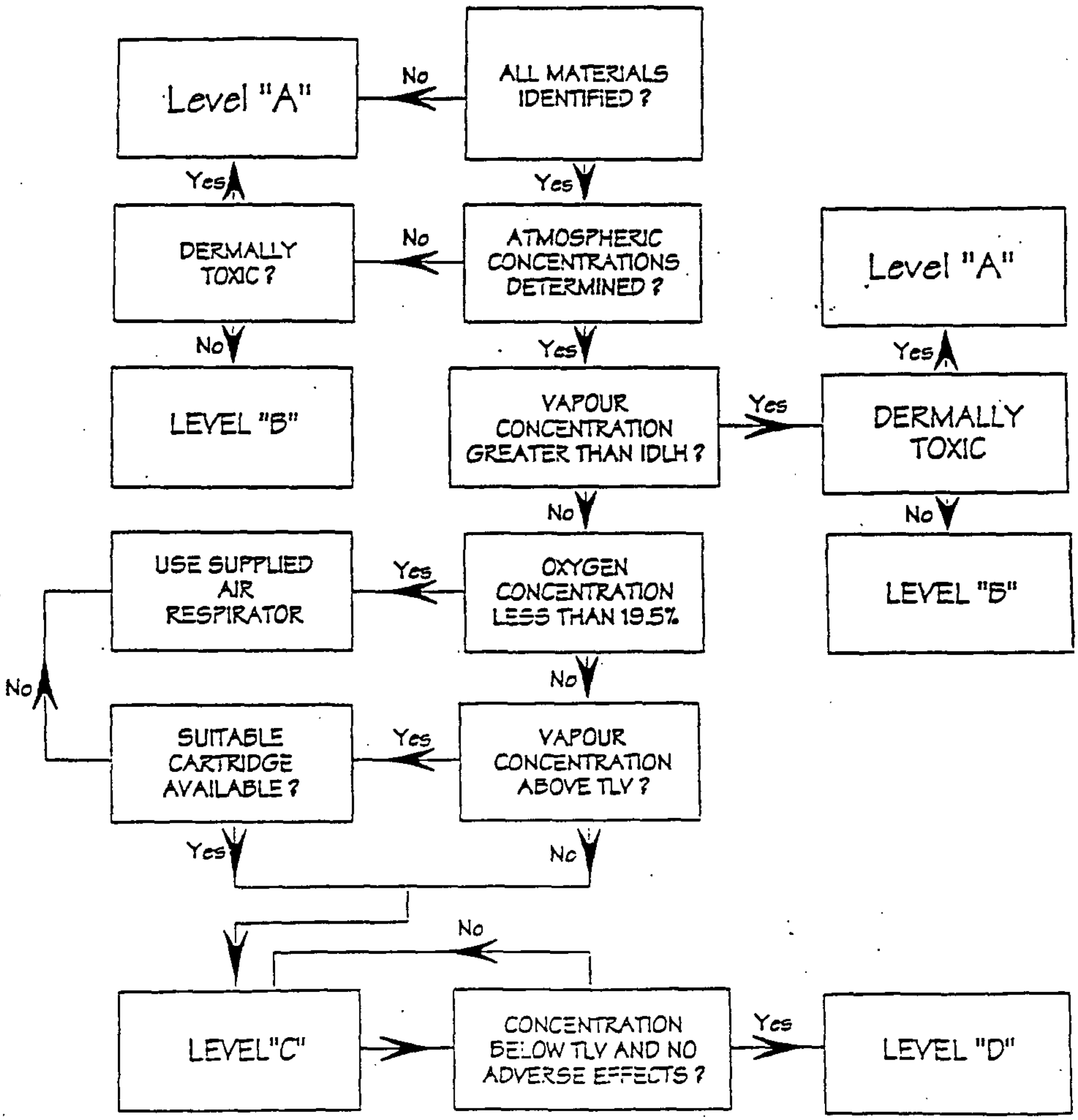
There are some precautionary signs that workers in PPE should be aware of which indicate a possible failure in their PPE. These include: discomfort, itching or irritation, detecting an odor, breathing difficulties, any discomfort, fatigue, nausea, increased pulse rate, chest pain, etc. If a worker experiences any of these symptoms he or she should leave the site immediately.

The decision of which level of PPE to wear is based upon achieving both the appropriate level of safety and the maximum ability to work. (Generally speaking, the higher the level of protection, the more awkward and cumbersome the PPE.) Figure 3 is a decision tree for selecting the appropriate level of PPE. Basically, the level of protection selected is based upon the following: the concentration of the chemical substance in the air and its toxicity; the potential for exposure to the chemicals; knowledge of chemicals on-site and an understanding of their properties such as toxicity and route of exposure.

TABLE 14: PROTECTION AND RECOMMENDED EQUIPMENT

US EPA Level	Recommended Equipment	Optional Equipment
A *Highest level of protection for skin,* eyes and respiratory system.	• Pressure-demand, full-facepiece self-contained breathing apparatus (SCBA) or pressure-demand supplied air respirator with escape SCBA • Fully-encapsulating, chemical-resistant suit • Inner chemical-resistant gloves • Chemical-resistant safety boots/shoes • Two-way radio communications	• Cooling unit • Coveralls • Long cotton underwear • Hard hat • Disposable gloves and boot covers.
B Highest level of respiratory protection, less skin protection as atmospheric contaminants on direct contact will not adversely affect any exposed skin.	• Pressure-demand, full-facepiece SCBA or pressure-demand supplied air respirator with escape SCBA • Chemical-resistant clothing (over-alls and long-sleeved jacket; hooded one- or two-piece chemical splash suit; disposable chemical-resistant one-piece suit) • Inner chemical-resistant gloves • Chemical-resistant safety boots/shoes • Hard hat • Two-way radio communications	• Coveralls • Disposable boot covers • Face shield • Long cotton underwear
C Same skin protection as level B but less respiratory protection as air contaminants have been identified and measured and a respirator cartridge to remove contaminant is available.	• Full-facepiece, air purifying, canister-equipped respirator • Chemical-resistant clothing (over-alls and long-sleeved jacket; hooded one- or two-piece chemical splash suit; disposable chemical-resistant one-piece suite) • Chemical-resistant safety boots/shoes • Hard hat • Two-way radio communications	• Coveralls • Disposable boot covers • Face shield • Escape mask • Long cotton underwear
D Some form of skin protection and no respiratory protection.	• Coveralls • Safety boots/shoes • Safety glasses or chemical splash goggles • Hard hat	• Gloves • Escape mask • Face shield

Figure 3. Decision Logic for Chemical Protective Clothing Selection



Respiratory Protection

A respirator functions to reduce adverse effects to the respiratory tract that result from breathing airborne contaminants. A respirator either removes the contaminants from the ambient air or supplies clean air to breathe. They are very important at a tire fire where the primary route of exposure is inhalation of particulates, fumes and vapors (see Section I).

There are two major classes of respirators: 1) air purifying respirators, which remove contaminants by passing the breathing air through a purifying element; and 2) an atmosphere supplying respirator, which provides a substitute source of clean breathing air. Self-contained breathing apparatus (SCBA) are an example of an atmosphere supplying respirator.

In the case of the Rhinehart Tire Fire (Winchester, Virginia), the National Institute for Occupational Safety and Health (NIOSH) recommended that the smoke plume from the fire should not be breathed. They stated that "if it is necessary for persons to work in the plume, then they should be provided with air-supplied respiratory protection and skin protection (level B)". In the literature regarding other tire fires, **level B was consistently the level of protection selected to protect workers on-site.**

The level of protection provided by PPE selection can be upgraded or downgraded based upon a change in site conditions or findings of investigations. When a significant change occurs, the hazards should be reassessed and changes made in the PPE as needed. Some indicators of the need for reassessment are: change in job tasks; change of season/weather; when temperature extremes or individual medical considerations limit the effectiveness of PPE; contaminants other than those previously identified are encountered; change in ambient levels of contaminants; change in work scope which effects the degree of contact with contaminants.

Heat Stress

Heat stress is one of the most common and potentially serious illnesses at sites where PPE is worn, and it is of particular concern at a tire fire site. Heat stress is caused by a number of interacting factors including environmental conditions, clothing, workload and the individual worker. Workers wearing PPE are susceptible to heat stress because the PPE limits the dissipation of the body's perspiration. The increased risk of excessive heat stress is directly influenced by the amount and type of PPE worn. When selecting PPE, each item should be carefully evaluated to assure its benefit is worth the increased risk of heat stress.

The major disorders associated with heat stress are heat cramps, heat exhaustion and heat stroke. These can range from mild conditions to very serious conditions to fatal. These conditions should be taken seriously and addressed immediately.

Several steps should be taken to reduce heat stress, including:

1) adjust work and rest schedules as needed;
2) provide shelter or shaded areas to protect personnel during rest periods;

3) maintain body fluids at normal level;
4) increase salt consumption;
5) wear personal cooling devices;
6) train workers to recognize and treat heat stress.

SITE CONTROL AND DECONTAMINATION

Site control and decontamination are as equally important as PPE for maintaining the health and safety of workers and also to help protect the health and safety of the surrounding population.

A site where there is a fire occurring must be controlled to minimize the possibility of exposure to chemicals and the transport of chemicals off-site. There are a variety of ways to reduce or eliminate exposure and transport. PPE is one method, several others include:

- setting up security and physical barriers to keep out unnecessary people;
- minimize the number of personnel and equipment on-site;
- establish work zones with controlled access points;
- implement appropriate decontamination procedures.

Work Zone Determination

It order to contain contamination and reduce or prevent exposure as much as possible, work zones should be set up. These zones typically include an "exclusion zone", a "contamination reduction zone" and a "support zone".

The exclusion zone is the innermost area surrounding the fire and it is generally contaminated. All people entering the exclusion zone must wear the determined level of protection. Factors that should be considered when setting up the exclusion zone are physical evidence of contamination, the distance needed to prevent the fire (or an explosion) from affecting personnel outside the zone, and the potential for contaminants to be blown from the area.

The contamination reduction zone is between the exclusion zone and the support zone. It provides an area for transition between the contaminated area and the clean area, and contains the decontamination stations.

The zone furthermost from the fire would be the support zone. It is considered the clean or non-contaminated area where support equipment (such as the command post, equipment trailer, etc.) is located. The location of the command post is dependent upon accessibility, wind direction and available resources.

The distances of each zone are based on site specific considerations, including room to conduct operations, weather conditions, terrain of the site, air contamination and dispersion calculations, clean up activities required, and the surrounding community.

Decontamination

Decontamination of personnel coming from the exclusion zone is critical to reducing exposure to contaminants. In the case of a tire fire, decontamination is very important because of the deposition and volatilization of the chemicals. A formal decontamination plan should be written as part of the emergency response plan. Generally speaking, decontamination methods are either physical or chemical. Physical methods include scraping, brushing, rinsing, vacuuming, heating, and cooling, while chemical methods include dissolving, neutralizing, using surfactants, solidification or sterilization. The plan should address the following issues:

- layout of decontamination site
- number of personnel required for decontamination stations
- roles and functions of decontamination team
- decontamination methods
- protection levels for decontaminating team
- water treatment and disposal methods
- contaminant treatment and disposal methods
- emergency decontamination procedures
- post-decontamination procedures (Hosty, 1992).

DISCUSSION

The chemicals that are released from a tire fire can have harmful health effects to people exposed, which could potentially include on-site workers and nearby residents. The proper selection and use of PPE is essential to assuring exposure is minimized. This would include using a level of protection that would offer sufficient skin and respiratory protection from exposure to the target chemicals. Given the toxicity and concentrations of the target chemicals within close proximity to the fire, it is generally recommended that workers use level B protection; however, a decision about proper level of protection should be made on a case-by-case basis. Factors such as wind direction and speed, geography, proximity to the fire, size of fire, and chemicals that are detected at the fire will all play a part in the decision regarding level of protection.

Safety issues such as site control and decontamination are also key to assuring exposure to on-site workers is minimized. There are several aspects to site control that should be implemented. Of most importance is determining and instituting work zones. Decontamination procedures are also critical to reducing exposure to on-site workers and other individuals (family members, workers in the support zone, media, etc.).

If PPE is used and proper site controls and decontamination procedures are implemented, they should minimize the exposure to the chemicals from a tire fire.

V. HEALTH FACT SHEETS

INTRODUCTION

A tire fire can provide a significant threat to the environment, the health of the personnel responding to the fire and potentially to nearby residents. In addition, extinguishing the fire can pose a major challenge to responders. However, with a well trained response team and effective emergency management, these challenges can be overcome. This is often followed by the greater challenge of communicating effectively with the media and citizens about the situation.

Generally, the public response to an incident such as a tire fire is one of alarm, fear, anxieties and misunderstandings. The public smells the tires burning, sees the black smoke plume, and emergency response personnel equipped with "moon suits". This picture does not convey a message of a safe, controlled situation. Consequently, it is imperative that effective communication with the media and public start immediately and continue until all objectives have been met and there is no longer interest or a need to communicate. Below is a general discussion about effectively communicating with the media and the public (assumed to be nearby residents). This discussion is followed by three health fact sheets that serve as templates that could be adapted and used during a tire fire. The fact sheets are for the media, the public, and on-site workers.

EFFECTIVE COMMUNICATION METHODS

<u>Media</u>

The media are there to obtain information about the incident (a story) and convey it to the public as soon as possible. If emergency site coordinators do not provide information to the media in a timely manner, they will find it elsewhere. It is a much better strategy to be prepared to talk with the media and give them the information they are seeking as soon as possible, than to let someone else function as the spokesperson.

The media are primarily interested in the who, what, where, when and how questions. Peter Sandman writes in "Explaining Environmental Risk" that "reporters were asked to specify which information they would need most urgently in covering an environmental risk emergency. Most reporters chose the basic risk information, saving the details for a possible second-day story. What happened, how it happened, who's to blame and what the authorities are doing about it all command more journalistic attention than toxicity during an environmental crisis."

When talking with the media keep the information simple, clear and concise. Reporters are not technically trained and typically do not understand chemical risk assessment jargon; therefore, they will not be able to convey the message effectively.

Another point regarding the media is that it can often be helpful in getting information out to the public. If the nearby residents need to be informed of the situation or some action that is going to be taken, the media can help distribute the information.

General Public

The public's reaction to an environmental hazard is not based upon the scientific facts; instead it is dependent upon characteristics such as familiarity with the hazard, whether or not the risk was encountered voluntarily, whether or not it is controllable, natural and memorable. These characteristics are part of how the public defines the risk of a situation. Appendix C lists eleven characteristics that help define the public's view of a situation's risk. It is important to understand people's perception about the risk of a situation (in this case, a tire fire) in order to communicate effectively about the risk. Peter Sandman has grouped these characteristics and calls them "outrage". He states that to the public, risk means outrage, not toxicity and probability. He also concludes that where the hazard is high, the job of risk communicators is to make outrage higher. Where the hazard is low, the job is not to explain that the risk is low but rather to keep the level of outrage low.

Controlling the level of outrage is not accomplished by simply following a few steps, rather it takes a thorough understanding of the public's concerns and the hazards of the situation. However, some ideas that can be pursued to control outrage are as follows:

- Share the power with the citizens. That is, consult with the nearby residents, form a citizens advisory committee, involve them in the policy decision process from the beginning.

- Keep the information flowing. If the fire was to burn for months, keep the citizens informed for the entire time. Make information readily accessible to them.

- Be open and honest about the situation. If a risk communicator does not have the answer to a question, he or she should acknowledge that fact. An honest and open approach will build a trust with the people and just as importantly, will build credibility.

HEALTH FACT SHEETS

Following this section are three health fact sheet templates that can be adapted and used in the case of a tire fire. The fact sheets are for the public, the media, and the on-site workers. Information pertaining to the site will need to be added to the fact sheets in order to address the specific needs of the site and situation. In the fact sheets a hypothetical tire fire at the Dorman pile in Pierce County is used as an example. As part of a tire fire contingency plan, it would be advisable to prepare fact sheets in advance, to be available for immediate distribution in the event of a fire. Maps detailing evacuation routes and/or probable impact areas may also be prepared in advance.

MEDIA HEALTH FACT SHEET

Press Release
Tacoma-Pierce County Health Department

DORMAN TIRE PILE FIRE

A fire is burning in the Dorman Tire Pile, located off Highway 702 near McKenna. This pile, on 11 acres, contains approximately 1.5 - 2 million tires. Pierce County Fire Department is responding to the fire, as well as the Washington State Department of Ecology, Pierce County Emergency Management and the Tacoma-Pierce County Health Department.

Tires are made primarily of rubber, oil, fillers, and fibers made from steel, nylon, polyester or rayon. A tire fire produces gases and vapors, oil runoff and airborne particles (particulates). Some of the chemicals that are found in these products include metals (lead, arsenic, barium), inorganic chemicals (sulfur dioxide, carbon monoxide, sulfuric acid) and hydrocarbons (aromatic and polyaromatic). A hydrocarbon is a chemical that contains both hydrogen and carbon. Some of the chemicals can cause cancer, affect the respiratory system or the central nervous system, or cause burns and irritation to the skin and eyes if a person is exposed to high enough levels for a long enough period of time.

Continuous air monitoring has been conducted since the fire started. Results indicate that the air close to the fire contains harmful levels of chemicals. This area has been restricted and only trained personnel wearing protective clothing and equipment can enter the area. Air samples have also been taken at the boundary of the site and at four locations in the surrounding community. Results from these samples show that the chemicals detected at the edge of the site and in the nearby community have not reached levels that cause a health concern.

As an extra safety precaution, the following recommendations are given:

1. Individuals living in the direction of the smoke plume minimize their time spent outdoors to avoid exposure to the smoke from the fire.
2. Persons with respiratory problems should take extra care to avoid breathing the smoke, to avoid aggravating their condition. Voluntary evacuation should be considered if a respiratory condition is being aggravated.
3. Any garden vegetables exposed to tire fire smoke should be washed thoroughly prior to eating.

For questions and further information contact the Tacoma-Pierce County Health Department at 591-6047.

PUBLIC HEALTH FACT SHEET

Tacoma-Pierce County Health Department

DORMAN TIRE PILE FIRE

A fire is burning in the Dorman Tire Pile, located off Highway 702 near McKenna. This pile, on 11 acres, contains approximately 1.5 - 2 million tires.

Tires are made primarily of rubber, oil, fillers, and fibers made from steel, nylon, polyester or rayon. A tire fire produces dark, sooty smoke, foul odors, and intense heat. Specifically, a tire fire produces gases and vapors, oil runoff and airborne particles (particulates). Some of the chemicals that are found in these products include metals (lead, arsenic, barium), inorganic chemicals (sulfur dioxide, carbon monoxide, sulfuric acid) and hydrocarbons (aromatic and polyaromatic). A hydrocarbon is a chemical that contains both hydrogen and carbon. The chemicals released from a tire fire can be harmful if a person is exposed to high enough levels for a long enough time. Some of the chemicals can cause cancer, affect the respiratory system or the central nervous system, or cause burns and irritation to the skin, eyes and nose.

Results from air samples show that the air very close to the fire contains chemical levels that are harmful to humans. This area has been restricted and only trained personnel wearing protective clothing and equipment can enter the area. Air samples have also been taken at the boundary of the site and at four locations in the surrounding community. Results from these samples show that the chemicals detected at the edge of the property are well below the level a worker can be safely exposed to eight hours a day. The results from the samples in the surrounding community showed there were only trace levels of chemicals. However, air monitoring will continue to assure the levels do not increase.

As an extra safety precaution, the following recommendations are given:

1. Individuals living in the direction of the smoke plume minimize their time spent outdoors to avoid exposure to the smoke from the fire.
2. Persons with respiratory problems should take extra care to avoid breathing the smoke, to avoid aggravating their condition. Voluntary evacuation should be considered if a respiratory condition is being aggravated.
3. Any garden vegetables exposed to tire fire smoke should be washed thoroughly prior to eating.

For questions and further information contact the Tacoma-Pierce County Health Department at 591-6047.

ON-SITE WORKERS HEALTH FACT SHEET

Tacoma-Pierce County Health Department

PIERCE COUNTY TIRE FIRE
RESPONSE PERSONNEL

This information is provided to individuals working in the vicinity of a tire fire to explain the health risks involved with exposure to tire fire by-products and procedures to be followed to minimize any risks. **Please read carefully.**

Tire fires produce foul odors, dense smoke, intense heat and a variety of hazardous by-products. Tires are made primarily of rubber, oil, fillers, and fibers made from steel, nylon, polyester or rayon. A tire fire produces gases and vapors, oil runoff and airborne particles (particulates). Some of the chemicals that are found in these products include metals (lead, arsenic, barium), inorganic chemicals (sulfur dioxide, carbon monoxide, sulfuric acid) and hydrocarbons (aromatic and polyaromatic). A hydrocarbon is a chemical that contains both hydrogen and carbon.

These chemicals can be harmful if a person is exposed to high enough levels for a long enough period of time. Some of the chemicals can cause cancer, effect the respiratory system or the central nervous system, or cause burns and irritation to the skin and eyes.

Exclusion zones have been established in areas considered hazardous, as indicated by air monitoring. It is important that proper protective equipment, as required by the on-scene commander, be worn at all times while in these areas. Exposure to tire fire ash and oil runoff in all areas should also be minimized through use of proper protective equipment.

Following work in the exclusion zone or contact with oil, ash, or smoke, proper decontamination procedures, as established on site, should be followed carefully.

Some short-term symptoms that could occur after exposure to the chemical emissions or intense heat include: eye, nose and throat irritation; breathing difficulties; burning sensation; lightheadedness; giddiness; or headache. If you experience any of these symptoms, or any other symptoms that you are concerned about, contact your supervisor immediately.

For questions and further information, contact your supervisor, or the Tacoma-Pierce County Health Department at 591-6047.

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APPENDIX A

TOXICOLOGICAL ASPECTS OF TARGET CHEMICALS

APPENDIX A : TOXICOLOGICAL ASPECTS OF TARGET CHEMICALS

*Hazard ratings were obtained from *Dangerous Properties of Industrial Materials* (Sax, 1979). Refer to that document for a complete discussion.

NAME OF SUBSTANCE	ACENAPHTHENE
CAS REGISTRY NUMBER	83-32-9
HAZARD RATING	
CLASSIFICATION	Suspected Carcinogen

Acenaphthene has been shown to cause cancer in animal studies. It is also an irritant to skin and mucous membranes.

NAME OF SUBSTANCE	ACENAPHTHYLENE
CAS REGISTRY NUMBER	208-96-8
HAZARD RATING	
CLASSIFICATION	Suspected Carcinogen

Acenaphthylene has been shown to cause cancer in animal studies.

NAME OF SUBSTANCE	ARSENIC
CAS REGISTRY NUMBER	7440-38-2
HAZARD RATING	3
CLASSIFICATION	Carcinogen

Arsenic is classified as a human carcinogen because of the increased lung cancer mortality in populations exposed to arsenic (primarily through inhalation) and because of the increased skin cancer incidence in several populations consuming drinking water with high arsenic concentrations. There have also been many cases of secondary carcinoma of internal organs, e.g., colon, bladder, gallbladder, pancreas, liver, ureter, prostate, lymph nodes, and bronchi as a consequence of metastases of primary skincancer induced by arsenic exposure. Arsenic induced cancer usually occurs at more than one site and has a long latent period, i.e., 13 to 50 years after the initial exposure. In industry, acute poisoning from solid arsenic is rare, although if swallowed arsenic can be an acute poison. Subacute and chronic poisoning usually results from exposure to arsenic containing dust and fume. Children appear to be more sensitive to arsenic than adults. It has been shown to be teratogenic, mutagenic and carcinogenic in animal studies.

NAME OF SUBSTANCE	BARIUM
CAS REGISTRY NUMBER	7440-39-3
HAZARD RATING	3
CLASSIFICATION	Irritant

Barium is considered to be highly toxic. Solid barium is poisonous when ingested. Symptoms are severe abdominal pain, vomiting, rapid pulse, paralysis of arms and legs and eventually coma and death. At lower concentrations the usual result is irritation of the eyes, nose, and throat, and of the skin, producing dermatitis.

NAME OF SUBSTANCE	BENZENE
CAS REGISTRY NUMBER	71-43-2
HAZARD RATING	3.
CLASSIFICATION:	Carcinogen

Benzene poisoning most commonly occurs through inhalation of vapors. It can also penetrate the skin. It can act as an irritant to the skin causing burning and in severe cases blistering. Exposure to high concentrations may result in acute poisoning and cause central nervous system effects, such as, dizziness, weakness, euphoria, headache, nausea, tightness in chest and staggering. High concentrations may also be irritating to the mucous membranes of the eyes, nose, and respiratory tract. Chronic exposure to benzene may include central nervous system effects but the major effect is on bone marrow cells. It has been shown to be a human carcinogen by inhibiting normal bone marrow functions. Following absorption of benzene, elimination is primarily through the lungs, when fresh air is breathed.

NAME OF SUBSTANCE	BENZ(a)ANTHRACENE
CAS REGISTRY NUMBER	56-55-3
HAZARD RATING	
CLASSIFICATION	Carcinogen

Benz(a)anthracene has been shown to be a weak carcinogen.

NAME OF SUBSTANCE	BENZO(a)PYRENE
CAS REGISTRY NUMBER	50-32-8
HAZARD RATING	3
CLASSIFICATION	Carcinogen

Benzo(a)pyrene has beenshown to be a powerful carcinogen through several different exposure routes. It has also been shown to cause teratogenic effects in animal studies.

NAME OF SUBSTANCE	BENZO(b)FLUORANTHENE
CAS REGISTRY NUMBER	205-99-2
HAZARD RATING	
CLASSIFICATION	Carcinogen

Benzo(b)fluoranthene has been shown to cause cancer in animal studies.

NAME OF SUBSTANCE	BENZYLCHLORIDE
CAS REGISTRY NUMBER	100-44-7
HAZARD RATING	3
CLASSIFICATION	Suspected Carcinogen

Benzylchloride has been shown to be a carcinogen and mutagen in animal studies. It is also a strong irritant to the eye and nasal passages and a moderate irritant when ingested.

NAME OF SUBSTANCE	BUTADIENE
CAS REGISTRY NUMBER	106-99-0
HAZARD RATING	3
CLASSIFICATION	Suspected Carcinogen

Butadiene has been shown to be a carcinogen and mutagen in animal studies. The vapors have also been shown to be an irritant to the eyes and mucous membranes. Inhalation of high concentrations can cause unconsciousness and death. If spilled on skin or clothing, can cause burns or frost bite (due to rapid vaporization).

NAME OF SUBSTANCE	CARBON MONOXIDE
CAS REGISTRY NUMBER	630-08-0
HAZARD RATING	3
CLASSIFICATION	Asphyxiant

Carbon monoxide is a colorless, odorless, tasteless gas. It is a chemical asphyxiant which means that it makes the blood incapable of transporting oxygen to the tissues. Carbon monoxide has an affinity for hemoglobin 210 times that of oxygen. Carbon monoxide is eliminated through the lungs when fresh air is inhaled. Symptoms of asphyxia start with shortness of breath and slight headache. This progresses into a severe headache, mental confusion and dizziness, impairment of vision and hearing, and eventually unconsciousness and death. Chronic effects as the result of repeated exposure at lower concentrations include heart irregularities, cerebral congestion, edema, and possibly mental or nervous system damage.

NAME OF SUBSTANCE	CARBON TETRACHLORIDE
CAS REGISTRY NUMBER	56-23-5
HAZARD RATING	3
CLASSIFICATION	Suspected carcinogen

Carbon tetrachloride has been shown to be a carcinogen and teratogen in animal studies. It is a skin and eye irritant and damages the central nervous system, respiratory functions and the gastrointestinal (GI) tract. Exposure to concentrations too low to cause unconsciousness, usually results in severe gastro-intestinal upset and may progress to serious kidney and hepatic damage. There is a lot of variation in individual susceptibility to carbon tetrachloride.

NAME OF SUBSTANCE	CHLORIDE
CAS REGISTRY NUMBER	
HAZARD RATING	
CLASSIFICATION	

The toxicity of chloride varies widely. Sodium chloride (table salt) has a very low toxicity, while carbonyl chloride (phosgene) can cause death in small doses.

NAME OF SUBSTANCE	CHLOROFORM
CAS REGISTRY NUMBER	67-66-3
HAZARD RATING	3
CLASSIFICATION	Suspected Carcinogen

Chloroform has been shown to be a carcinogen and teratogen in animal studies. It is also an irritant to the eye. When inhaled it effects the eyes. It has been used as an anesthetic, but this is being abandoned due to its toxic effects. When inhaled, it initially gives a feeling of warmth, then it irritates the mucous membranes and skin. Prolonged inhalation will cause paralysis accompanied by cardiac respiratory failure and death.

NAME OF SUBSTANCE	CHROMIUM
CAS REGISTRY NUMBER	7440-47-3
HAZARD RATING	3
CLASSIFICATION:	Carcinogen

Some forms of chromium have been shown to cause lung, nasal cavity, paranasal sinus, stomach and larynx cancer in both human and animal studies. Chromium has a very high respiratory toxicity rating, as well as having a corrosive action on the skin and mucous membranes. Ulcers form around the base of the nails, on the knuckles, hands and forearms after exposure. The ulcers heal slowly and are virtually painless. It can also cause eczematous dermatitis.

NAME OF SUBSTANCE	CHRYSENE
CAS REGISTRY NUMBER	218-01-9
HAZARD RATING	3
CLASSIFICATION	Suspected Carcinogen

Chrysene is a coal tar derivative. It has been shown to cause cancer in animals. It is highly toxic for dermal and inhalation routes.

NAME OF SUBSTANCE	COAL TAR PITCH
CAS REGISTRY NUMBER	8007-45-2
HAZARD RATING	3
CLASSIFICATION	Carcinogen

Coal tar pitch is the black or dark brown amorphous residue that remains after the redistillation process. The volatiles contain a large quantity of lower molecular weight polycyclic hydrocarbons (naphthalene, fluorene, anthracene, acridine, phenanthrene). As these hydrocarbons volatilize into the air there is an increase of the higher weight polycyclic aromatic hydrocarbons (benzo(a)pyrene). These are the ones that are known to be carcinogenic. Air polluted with coal tar pitch has large amounts of phenanthrene, anthracene, pyrene, and carbazole. The composition of the hydrocarbons varies with time and distance, suggesting differences in stability of the compounds.

NAME OF SUBSTANCE	CUMENE
CAS REGISTRY NUMBER	92-82-8
HAZARD RATING	3
CLASSIFICATION	Irritant

Cumene is a potent irritant when inhaled. It depresses the central nervous system and has a potent narcotic action. It takes some time for it to have an effect, however, the effects last a long time. It is also considered a skin and eye irritant. There is some indication that cumene has a greater acute toxicity than benzene or toluene.

NAME OF SUBSTANCE	DIBENZ (a,h) ANTHRACENE
CAS REGISTRY NUMBER	53-70-3
HAZARD RATING	
CLASSIFICATION	Carcinogen

A powerful carcinogen in animal studies.

NAME OF SUBSTANCE	1,2-DICHLOROPROPANE (propylene dichloride)
CAS REGISTRY NUMBER	78-87-5
HAZARD RATING	2
CLASSIFICATION	Mutagen

1,2-Dichloropropane has been shown to cause mutagenic effects in animals studies. It is considered one of the more toxic chlorinated hydrocarbons. It is an irritant to the skin and eyes.

NAME OF SUBSTANCE	ETHYLENE DICHLORIDE (1,2 dichloroethane)
CAS REGISTRY NUMBER	107-06-2
HAZARD RATING	3
CLASSIFICATION	Suspected Carcinogen

Ethylene Dichloride has been shown to cause cancer and mutagenic effects in animal studies. When heated, ethylene dichloride emits very toxic fumes. When inhaled it effects the central nervous system and when ingested it effects the gastro-intestinal tract. It is a strong irritant to the eyes and skin, and has a strong narcotic effect. Acute exposure to high concentrations results in irritation of the eyes, nose and throat, followed by dizziness, nausea, vomiting, increasing stupor, cyanosis, rapid pulse, and loss of consciousness. It is toxic to the liver and kidney.

NAME OF SUBSTANCE	FLUORIDE
CAS REGISTRY NUMBER	
HAZARD RATING	3
CLASSIFICATION	

Inorganic fluorides are generally highly irritating and toxic. Their effects can be combined into three groups: 1) acute symptoms are intense irritation of the skin, eyes, and mucous membranes, through both inhalation and ingestion; 2) acute systemic intoxication, usually through ingestion; and 3) chronic exposure effects the bones and can result in mottled teeth and crippling skeletal abnormalities. Large doses can cause very severe nausea, vomiting, diarrhea, abdominal burning and cramp-like pains. It can also cause aggravated attacks of asthma.

NAME OF SUBSTANCE	HEXACHLOROETHANE
CAS REGISTRY NUMBER	67-72-1
HAZARD RATING	3
CLASSIFICATION	Suspected Carcinogen

Hexachloroethane has been shown to cause cancer in animal studies. Exposure also results in liver damage. When heated it emits highly toxic fumes of phosgene. Moderate to high toxicity through oral and dermal routes of exposure.

NAME OF SUBSTANCE HEXANE
CAS REGISTRY NUMBER 110-54-3
HAZARD RATING 2
CLASSIFICATION

Low toxicity through ingestion. It is used as a food additive for humans. It can cause a motor neuropathy in exposed workers. It can cause irritation to the respiratory tract and act as a narcotic in high concentrations. Exposure through inhalation can result in drowsiness, fatigue, loss of appetite, and irritation to the eye. Dermal exposure can cause itching, blister formation, pigmentation and pain.

NAME OF SUBSTANCE LEAD
CAS REGISTRY NUMBER 7439-92-1
HAZARD RATING 3
CLASSIFICATION: Carcinogen

Inhalation exposure to lead fumes can result in significant accumulation of lead if the concentrations and exposures are high and frequent enough. Lead is a suspected human carcinogen of the lungs and kidney. Animal studies have shown it to be carcinogenic, however, human studies have not confirmed this finding. It has also been shown to have teratogenic effects in animal studies. Lead also effects the human central nervous system and is a moderate irritant. Lead can be absorbed through the skin.

NAME OF SUBSTANCE METHYLENE CHLORIDE
CAS REGISTRY NUMBER 75-09-02
HAZARD RATING 3
CLASSIFICATION Suspected Carcinogen

Methylene chloride has been shown to be a carcinogen, teratogen, and mutagen in animal studies. It is very dangerous to the eyes. Other than inducing narcosis, it has very few other acute toxicity effects. It is a strong narcotic and effects the central nervous system. It can also cause dermatitis, nausea and accelerated pulse. It volatilizes easily.

NAME OF SUBSTANCE NICKEL
CAS REGISTRY NUMBER 7440-02-2
HAZARD RATING 3
CLASSIFICATION: Carcinogen

Many nickel compounds are carcinogenic in the form of respirable dusts and aerosols. The most common effect resulting from exposure to nickel compounds is the development of the "nickel itch", a type of dermatitis. There is a large variability in individual susceptibility to the dermatitis. It occurs more frequently under conditions of high temperature and humidity, when the skin is moist and chiefly affects the hands and arms.

NAME OF SUBSTANCE	PHENOL
CAS REGISTRY NUMBER	108-95-2
HAZARD RATING	3
CLASSIFICATION	Suspected Carcinogen

Phenol has been shown to be carcinogenic and mutagenic in animal studies. It is an irritant to the eyes and skin. Acute exposure results in central nervous system effects. Absorption through the skin is the primary route of exposure and can be very quick, resulting in headache, dizziness, irregular breathing, ringing in the ears, collapse and eventually death.

NAME OF SUBSTANCE	STYRENE
CAS REGISTRY NUMBER	100-42-5
HAZARD RATING	3
CLASSIFICATION	Suspected Carcinogen

Styrene has been shown to be a carcinogen and mutagen in animal studies. It is a skin and eye irritant, causing violent itching of the eyes and resulting in severe eye injury. It effects the central nervous system and its less than fatal toxic effects are usually transient and result in irritation and possible narcosis.

NAME OF SUBSTANCE	SULFUR DIOXIDE
CAS REGISTRY NUMBER	7446-09-5
HAZARD RATING	3
CLASSIFICATION	Mutagen, Irritant

Sulfur dioxide is an eye irritant. It has been shown to be a mutagen in animal studies. It is a suffocating and poisonous gas when inhaled. It will react with water or steam to produce toxic and corrosive fumes. It mainly affects the upper respiratory tract and bronchi.

NAME OF SUBSTANCE	SULFURIC ACID
CAS REGISTRY NUMBER	7664-93-9
HAZARD RATING	3
CLASSIFICATION	Irritant

Sulfuric acid is an extremely strong irritant and is corrosive and toxic to tissue. Contact with the body results in rapid destruction of tissue and causes severe burns. It is a very dangerous irritant to the eye. There are no systemic effects from chronic ingestion of small amounts. It reacts with water or steam to produce heat and when heat decomposes the acid it emits highly toxic fumes. Contact can cause dermatitis. Repeated or prolonged inhalation of mists can cause chronic bronchitis, erosion of teeth and a very painful skin necrosis.

NAME OF SUBSTANCE	TOLUENE
CAS REGISTRY NUMBER	108-88-3
HAZARD RATING	3
CLASSIFICATION	Mutagen

Toluene is a derivative of benzene but does not effect the bone marrow like benzene does. It has been shown to cause mutagenic effects in animal studies. The major route of entry is inhalation. It can cause mild upper respiratory tract and eye irritation, as well as impaired coordination and reaction time at lower concentrations. It is a central nervous system depressant. Chronic exposures may lead to liver and kidney effects.

NAME OF SUBSTANCE	1,1,2-TRICHLOROETHANE
CAS REGISTRY NUMBER	79-00-5
HAZARD RATING	3
CLASSIFICATION	Suspected Carcinogen

1,1,2-Trichloroethane has narcotic properties and acts as a local irritant to the eyes, nose and lungs. It has been shown to cause cancer and mutagenic effects in animal studies. It may be dangerous to the liver and kidney. When heated it gives off very toxic fumes.

NAME OF SUBSTANCE	TRICHLOROETHYLENE
CAS REGISTRY NUMBER	79-01-6
HAZARD RATING	3
CLASSIFICATION	Suspected Carcinogen

Trichloroethylene has been shown to be a carcinogen, mutagen and teratogen in animal studies. It is a strong skin and eye irritant. Prolonged inhalation causes headache and drowsiness. Chronic exposure results in damage to liver and kidney.

NAME OF SUBSTANCE	VANADIUM
CAS REGISTRY NUBMER	1314-32-1
HAZARD RATING	3
CLASSIFICATION	Poison, Mutagen

Vanadium has been shown to cause mutations in animals. It is a very toxic chemical and is a poison when inhaled, ingested or absorbed. Acute effects include irritation to the respiratory system and the eyes.

NAME OF SUBSTANCE O-XYLENE
CAS REGISTRY NUMBER 95-47-6
HAZARD RATING
CLASSIFICATION

Ingestion of xylene will cause severe gastrointestinal distress. Inhalation results in chemical pneumonitis, pulmonary edema and hemorrhage. Direct eye exposure can result in corneal burns and conjunctivitis. Repeated, prolonged exposure may produce central nervous system excitation and then depression and possible liver and kidney damage.

NAME OF SUBSTANCE ZINC
CAS REGISTRY NUMBER 7440-66-6
HAZARD RATING 3
CLASSIFICATION: Carcinogen

Many zinc salts are known or suspected carcinogens. Generally speaking the zinc compounds are of low toxicity. The inhalation of high concentrations of zinc chloride fumes has been fatal due to lung damage. At lesser doses it can cause damage to the mucous membrane of the nasopharynx and respiratory tract. Soluble salts of zinc can cause nausea and vomiting in small doses.

APPENDIX B

FIELD SAMPLING DATA SHEETS
CHAIN-OF-CUSTODY FORMS

APPENDIX C

CHARACTERISTICS OF RISK TO THE PUBLIC

APPENDIX C: CHARACTERISTICS OF RISK TO THE PUBLIC

1. The distinction between risks that are voluntary and those which are coerced. This is as true for communities as for individuals. Coercion is seen as an assault which creates risk to those receiving the force.

2. A natural vs. a man-created risk. Radon is viewed as an act of God. A tanker oil spill is seen as an act of man and triggers greater outrage.

3. The familiar vs. the exotic or unfamiliar. Radon collects in the basement, a familiar location. Watching on TV an EPA cleanup by men in spacesuits creates a greater level of anxiety.

4. Not memorable vs. memorable. Memorable relates to how easy it is to conjure up thoughts of how bad things could be or how things could go wrong. Non-memorable relates to those situations we have lived in without a sense of apprehension. Memorable reactions can be stimulated by imagination or associations as well as real events. Talk of building waste incinerators conjures up on many minds thoughts of belching industrial smokestacks.

5. Not dreaded vs. a sense of dread. AIDS and cancer cause dread. A number of AIDS or cancer deaths will cause more alarm in a community than an equal number of deaths from other diseases. A 55-gallon drum marked "materials" will cause less anxiety than the same drum and content marked "waste".

6. Knowable vs. not knowable. Knowability relates to a number of factors such as scientific knowledge, familiarity, and detectability. Beliefs also may be treated as fact.

   Debate as to what is knowable may lead to conflict. Fear and outrage may increase as debate develops regarding what a risk level really is. The scientific capacity to detect minute amounts of substances increases this debate. Rates in discussing risk assessment takes a different meaning from rates in standard settings.

   An agency arguing that views of risk are exaggerations may be viewed by the public as disinterested and thereby increase fear and outrage.

7. Controlled by the individual (self) vs. controlled by others. Control is determined by who implements. Visualize the difference between sharpening the butcher knife and cutting the rib roast vs. holding the roast while someone else cuts. In many conflicts between government agencies and communities, the communities view themselves as holding the meat while the agency wields the knife.

   An agency sends mixed, contradictory messages when it tries to control or impose and at the same time try to say "don't worry". An agency which has a history of adverse impact on communities and people must acknowledge that history if it is to reduce dread and other anxiety-producing feelings.

Being knowledgeable is a part of the sense of control and voluntary participation. For an agency to share control means that it must allow the community to implement key controlling factors such as doing the study to determine need, desirability, et al.

8. Fair vs. unfair. One element of public perception of fairness is the perception as to distribution of risks and benefits. The belief is that benefits should benefit those sharing the risk and not skip the risk-bears. For instance, outrage may increase over a garbage incinerator proposal which includes bringing garbage from outside the area. Public perception of fairness may also have a "my risk vs. your risk" element.

9. Morally irrelevant vs. morally relevant. Developing public values that some factor, for example, environmental contamination, is wrong--EVIL--gives moral relevance to the issue. However, some morally relevant issues have less tradeoffs. The more incensed the public becomes, the higher its sense of risk. When the public becomes very angry, it may demand a zero-risk solution. If the public agency shows no signs of getting to zero risk, the public becomes angry again.

10. Trusted vs. not trusted. Health agencies are trusted by the public when they deliver a warning, when they say that something is dangerous. ("They are the experts.") The health agency may not be trusted when it says something is safe. (Others are saying it is not safe.) Distrust may be one of the community's chief measures of the size of risk.

11. Good process vs. bad process The processes of interaction between the governmental agency and the community may contribute to the community's sense of risk.

    A. Openness vs. secrecy. Openness reduces a public sense of risk. Agencies should tell what they will do and what happened. The public is more interested in being told than in what they are told.

    B. Acknowledge prior misdeeds vs. denial and stonewalling. The American public is tolerant of repentant sinners but distrusts those who legalistically deny that sin has occurred. "We goofed," is a necessary defense for public agencies.

    C. Caring communication vs. jargon and argument. What the speaker termed "technobabble" is distancing language.

Taken from "Hazard vs. Outrage: How the Public Sees Health Risk". Presentation by Peter M. Sandman, Ph.D., Rutgers University to the ASTHO Annual Meeting, April 6, 1989, Vail Colorado.

APPENDIX D

FIELD REFERENCE SHEET

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PARAMETER	SAMPLING EQUIPMENT	ACTION LEVEL
Direct Reading:		
Carbon Monoxide	Direct Reading Monitor or Detector Tube	25 mg/m^3
Combustible Gases	Combustible Gas Indicator	10% LEL
Organic Vapors	TIP Meter or HNU	---
Oxygen	Oxygen Monitor	19.5%
Sulfur Dioxide	Detector Tubes	2 mg/m^3
Sulfuric Acid	Detector Tubes	1 mg/m^3
Air Sampling:		
PAHs	37mm Teflon filter with air pump Washed XAD-2 Sorbent tube with air pump	---
VOCs	Tenax (R) tube with air pump	---